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Review

Large-scale nuclear and plastid phylogenomic analyses inform an updated Angiosperm Phylogeny Group classification: APG V

The Angiosperm Phylogeny Group

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Abstract We present here a revision of the APG classification that considers the extensive recent analyses of hundreds of nuclear and plastid genes for many angiosperm species. Although previous versions of the APG classification were largely based on uniparentally inherited markers (plastid DNA, typically maternally inherited), there has emerged since APG IV extensive evidence of widespread hybridization and inheritance of ancestral polymorphisms. Despite this evidence of gene-tree discordance, most of the APG IV classification is supported by both plastid and nuclear analyses, and only a few revisions are required in this update to make the classification parallel phylogenetic results in the literature. At the ordinal level, few changes in circumscription are required relative to the last APG version, although the fabids now comprise only the four nitrogen-fixing orders (Cucurbitales, Fabales, Fagales, and Rosales) and the malvids now include the former COM clade (Celastrales, Malpighiales, and Oxalidales), which, in nuclear DNA analyses, is no longer monophyletic. Oncothecales (with only Oncothecaceae) and Cardiopteridales (with Cardiopteridaceae and Stemonuraceae) are newly recognized, and Icacinales are restricted to Icacinaceae. Restriction of Aquifoliales and Bruniales to just Aquifoliaceae/Helwingiaceae and Bruniaceae, respectively, is also proposed. Huaceae and Columelliaceae are unplaced to order among the malvids and campanulids, respectively. At the family level, expanded circumscriptions include Tecophilaeaceae (including Ixioliriaceae, Asparagales), Frankeniaceae (including Tamaricaceae, Caryophyllales),

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Phytolaccaceae (including Agdestidaceae and Sarcobataceae, Caryophyllales), Helwingiaceae (including Phyllonomaceae, Aquifoliales), Gesneriaceae (including Calceolariaceae and Peltantheraceae, Lamiales), Pentaphylacaceae (including Sladeniaceae, Ericales), and Orobanchaceae (including Mazaceae, Paulowniaceae, Phrymaceae, and Wightiaceae, Lamiales). In Santalales, the limits of Santalaceae and Olacaceae are revised, and Erythropalaceae and Strombosiaceae are recognized as distinct from Olacaceae; Balanophoraceae are maintained as distinct from Santalaceae, but this is a tentative placement. Generic relationships in Dioscoreales require more analyses before a revised family classification can be proposed, so we maintain the APG IV families of this order.

Key words: angiosperms, classification, molecular systematics, nuclear DNA, organellar DNA.

1 Introduction

The goal of modern biological classification is to translate phylogenetic knowledge into an explicit system for communicating about biological taxa, and the challenges involved in conveying incomplete and imperfect knowledge through a universal system are faced by systematists working across the tree of life. Moreover, competing classifications for a group of organisms can impede communication and stifle use of any of these. Thus, the issues of which taxa to recognize and name—and the criteria governing these decisions—are common to classification of all organismal groups, and successful strategies and methods developed for one group may be applicable to other groups. Here, we present an updated version of the Angiosperm Phylogeny Group (APG) classification, a system first published almost three decades ago with explicit criteria and a community-based approach to classification. The principle of generating a classification through developing consensus of many specialists was a new concept at that time in the field of classification, which by and large had single authors, especially in the last 100 years. APG (1998) was the first such classification to involve numerous authors and broad community input and has served as an exemplar for many other fields (e.g., Pteridophyte Phylogeny Group, 2016); now, APG V provides a framework for incorporation of taxonomically broad studies of nuclear genes as the basis for a revised classification of angiosperms, a model that other groups may wish to follow.

Previous classifications of flowering plants from the Angiosperm Phylogeny Group (APG I–IV; 1998, 2003, 2009, 2016) relied almost exclusively on molecular phylogenetic studies of DNA regions from the plastid genome, first *rbcL* and then *atpB* (Chase et al., 1993; Savolainen et al., 2000a, 2000b). Plastid markers were chosen because there are many plastids per cell with a single genome per plastid, and they have a rate of molecular evolution appropriate for phylogenetic inference of higher-order relationships (Palmer, 1987; Ritland & Clegg, 1987). The availability of free, universal *rbcL* PCR and sequencing primers also made it inexpensive and easy to amplify and sequence this gene (Zurawski & Clegg, 1987). In contrast, most nuclear genes were difficult to sequence at that time, resulting in a focus on the highly repeated nuclear ribosomal RNA genes, which were included in some early studies (e.g., Hamby & Zimmer, 1992; Soltis et al., 1997). However, ribosomal trees showed lower resolution and support than the plastid trees, given the slower rate of evolution of ribosomal DNA (as opposed to the more rapidly evolving spacers such as nuclear ribosomal ITS). Because these early nuclear phylogenetic trees were largely congruent with inferences based on plastid genes (Nickrent & Soltis, 1995; Kuzoff et al., 1998), they were often combined with plastid gene sequences (e.g., Soltis et al., 2000). Ultimately, a 17-gene data set (Soltis et al., 2011) in concert with other published analyses, many involving numerous plastid (Yang et al., 2014; Sun et al., 2016) and mitochondrial (Zhu et al., 2007) genes, formed the backbone tree compiled for APG IV (2016).

Although inherited as a single unit with phylogenetically negligible levels of recombination (Doyle, 2022; but see Li et al., 2025, for a rare exception), plastid genomes (and to a lesser extent mitochondrial genomes) are nevertheless a rich source of phylogenetic information. However, we have long dealt with the reality that organellar genomes effectively mask evidence of hybridization and other potentially confounding processes (e.g., incomplete lineage sorting, ILS). Compounding this problem was that, historically, the main sources of nuclear gene trees were those of nuclear ribosomal RNA/DNA, which are generally subject to concerted evolution (Hills & Dixon, 1991) and thus also concealed evidence of reticulation.

Although the general congruence of plastid, mitochondrial, and nuclear topologies has long been noted, it has been recognized that ultimately, the APG classification would need to encompass results from an extensive sampling of nuclear genes (APG, 2016). During the past decade, the rapid accumulation of large data sets obtained from transcriptomes (e.g., Zeng et al., 2014, 2017; One Thousand Plant Transcriptomes Initiative, 2019; Yang et al., 2020; Zhang et al., 2020; Xiang et al., 2024), target sequence capture (e.g., Baker et al., 2022; Zuntini et al., 2024; Kates et al., 2024), and complete or near-complete genome assemblies (e.g., Guo et al., 2021; Kress et al., 2022; Lewin et al., 2022; Darwin Tree of Life Project Consortium, 2022) has provided a broad sampling of nuclear genes across angiosperms. These massive and rapidly growing nuclear data sets have brought us to a pivotal moment in the history of APG: a shift toward reliance on the nuclear genome to inform classification. In general, congruence of analyses of large numbers of nuclear and organellar genes provides greater confidence in proposing a revised classification, and if incongruence is detected, examination of nuclear genes is essential for understanding the evolutionary processes behind this discordance.

As a general summary of the novelties in this update of APG, we have observed well-supported relationships in multiple nuclear gene trees (e.g., Zuntini et al., 2024) that mostly agree with previous plastid trees (e.g., Li et al., 2021; reviewed in Zhang & Ma, 2024), despite the presence of incongruence and evidence of deep reticulation at various points (Stull et al., 2020, 2023; Zuntini et al., 2024). We wish to make it clear that, as with all previous APG iterations, APG V is a summary of published studies and is not based on any single or set of available analyses, and the tree in Fig. 1 is a

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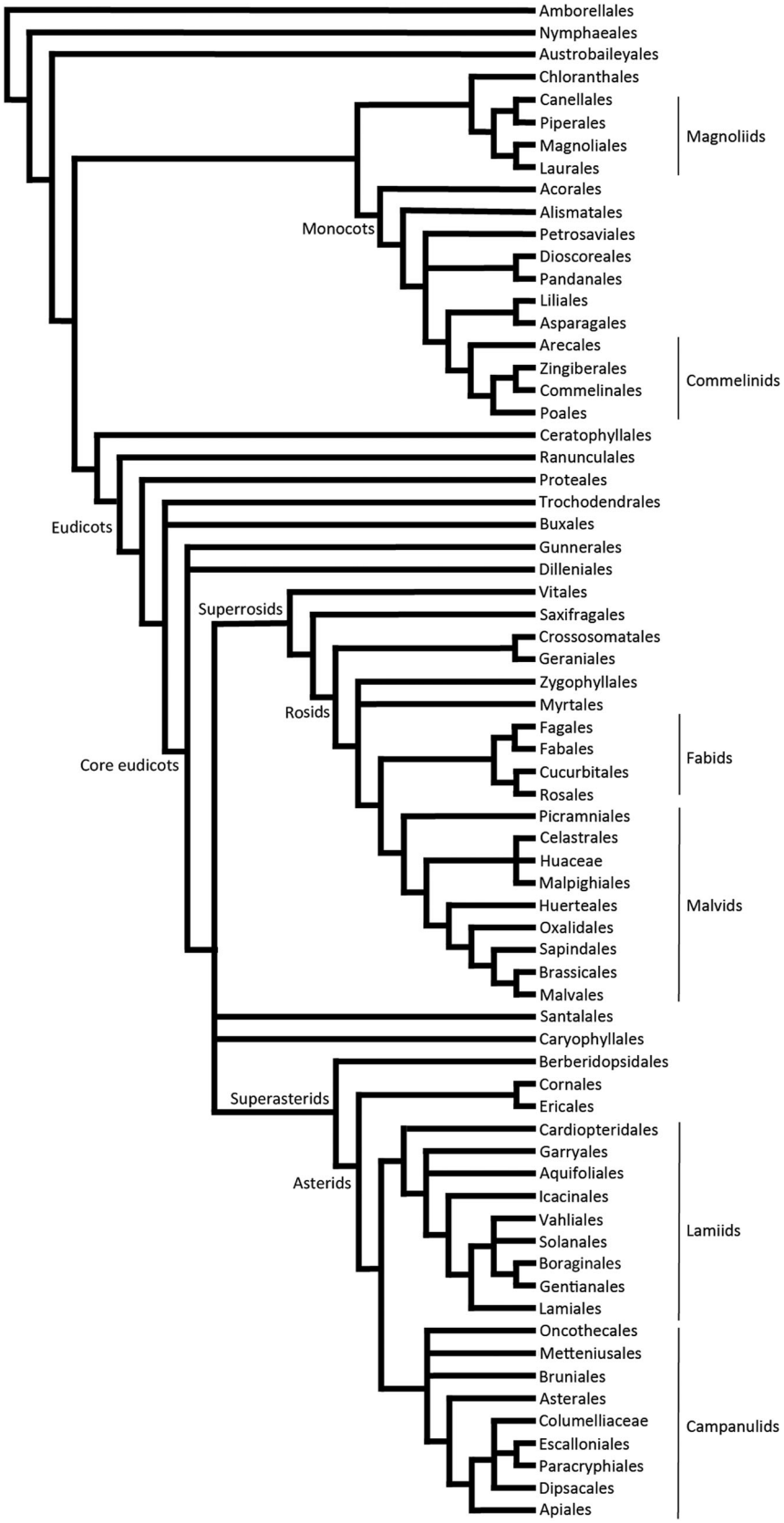


Fig. 1. Summary APG V tree showing relationships of orders based on nuclear gene sequences (e.g., One Thousand Plant Transcriptomes Initiative, 2019; Stull et al., 2020; Yang et al., 2020; Xiang et al., 2024; Zuntini et al., 2024).

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synopsis and therefore not based on an analysis of a specific DNA data set. However, before moving on to changes needed in APG IV to produce an update (APG V), we first revisit the two general principles under which APG has operated.

In addition to the APG classification of angiosperms, there have been similar recent efforts, for example, that of the Bryophyte Phylogeny Group (BPG; Li et al., 2024) and the Pteridophyte Phylogeny Group for lycopods and ferns (PPG, 2016). Both, however, have been almost entirely based on plastid DNA data, and in this respect, APG is leading the efforts to incorporate the large amount of nuclear gene data being produced during the "genomic era." Similar efforts have been made in several angiosperm families, for example, the Legume Phylogeny Working Group (LPWG; 2017), the Grass Phylogeny Working Group (GPWG, 2001, 2012, 2025), and orchids (Chase et al., 2015), among many.

2 General Principles of the APG Classification (a reprise)

The authors of the previous APG classifications have recognized that multiple classifications may be derived from published evidence, and the choice of which to accept has always depended on the author′s taxonomic philosophy and approach. There is more than one way to cut up a phylogenetic tree, and authors have varied opinions on how best to do this to produce a classification. In APG (1998), the authors explained the process used to determine the number of orders to recognize and the criteria that were considered important. APG (1998) previously recognized the families and orders that had been in the prevailing morphologically based classifications of that time to provide stability and familiarity, but these did not always have the traditional composition. This task was made easier because in the 1970s, substantial agreement had developed among the systems being proposed. However, the authors of APG (1998) made a conscious effort to minimize the number of orders to keep the system as simple as possible. Thus, the general arrangements of the monocot orders and families were those proposed by Dahlgren et al. (1985). The original APG (1998) for all angiosperms was similar in structure to that proposed by Dahlgren (1975), which was notable in the use of morphological cladistics to provide a picture of angiosperm relationships. Dahlgren (1975) recognized roughly 30 superorders, which is similar in number and names to the 40 orders in APG (1998). This general structure has been maintained in all subsequent APG iterations, which have merged traditional concepts with the results of modern DNA analyses informed by the input of experts. To maintain stability and mediate diverse opinions when changes have been proposed, a set of general guiding principles was established, which we reiterate below.

2.1 Monophyly

Monophyly has been the most important principle since the first version of APG (1998; published without a number). Before this, paraphyly and polyphyly in angiosperm classifications based on morphology were addressed by Judd et al. (1994), who proposed a series of revised family limits for cases of non-monophyly, which were incorporated into APG (1998). Other cases of paraphyly, that is, Amaranthaceae/Chenopodiaceae, Boraginaceae/Hydrophyllaceae, and Sapindaceae/Aceraceae/Hippocastanaceae, were resolved in APG (1998). Since then, circumscriptions of several other problematic families have been substantially revised to accommodate new results of molecular phylogenetic studies, for example, Acanthaceae, Aristolochiaceae, Lamiaceae, Orobanchaceae, Scrophulariaceae, and Verbenaceae (APG IV, 2016). At the ordinal level, there have been fewer problems with non-monophyly; however, given that the APG orders were mostly based on DNA results and unique to APG, such problems should not have been expected.

Monophyly as a guiding principle becomes more complicated when faced with evidence of species/clades created by hybridization, generating numerous instances of gene-tree discordance that led to a lack of clear evidence of relationships (Jeffroy et al., 2006; Steenwyk et al., 2023). Speciation events are real and have been empirically demonstrated (reviewed in Hernández-Hernández et al., 2021). However, some genomes have been shown to be "porous," with a few loci subject to extensive gene flow (Lexer et al., 2009; Campbell et al., 2018), whereas most genes in these same species do not show gene flow. We must, therefore, expect to encounter discordant gene trees in some groups despite no other evidence of hybridization. Nonetheless, some tests may help to determine which gene trees have signs of incomplete lineage sorting (also referred to as hemiplasy, deep coalescence, retention of ancestral polymorphisms, or trans-species polymorphisms) or introgression (e.g., resulting from horizontal gene transfer or hybridization; Smith et al., 2020; Morales-Briones et al., 2021; Singhal et al., 2021; Komarova & Lavrenchenko, 2022), the former being unreliable for phylogeny reconstruction and the latter being directly responsible for gene-tree discordance.

Addressing reticulation in classification is problematic at any level, but despite evidence of hybridization in the ancestry of some older clades (recognized as higher taxa, such as orders and families), these clades are generally monophyletic, even though we expect that in some gene trees, they may be otherwise. Taxa with a hybrid origin can be named on this basis (e.g., Knapp et al., 2004, for allotetraploids in *Nicotiana*, Solanaceae). We do not propose to assign taxa of hybrid origin to the same taxon as one of their parents, even if the hybrid genome has retained more genes from that parent. Instead, we recommend naming them in the same manner as any family or order, but note their putative hybrid origin (e.g., as we have for Ericales in this classification, which appears to be hybrid in origin; Stull et al., 2020). However, understanding more general patterns of diversification when both parental and hybrid clades are included is less straightforward, but purely for the purposes of a classification, understanding the causes for incongruence is not strictly essential. Other naming strategies are possible (e.g., use of the PhyloCode approach; de Queiroz et al., 2020), but these are beyond the scope of the APG classification.

We now know that there is substantial nuclear gene-tree and nuclear-plastid discordance for interrelationships of some APG orders and other informally named clades (e.g., demise of the COM clade), but at present, we cannot ascertain their parentage or other potential processes leading to incongruence (e.g., paralogy, poor taxonomic

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sampling, or gene-tree discordance due to population processes). It is important to reiterate that, although hybridization and ILS appear to be widespread in most groups of flowering plants, gene-tree discordance affects only a few of the taxonomic categories on which APG is focused (as discussed below).

2.2 Stability

The APG stability principle states that if a currently accepted APG family or order has been shown to be monophyletic, it will not be subsequently split. Although not an issue in the first two iterations of APG (1998, 2003), stability soon became nearly as important as monophyly. Incorporation of DNA sequence data into classification has in some cases, changed names and families of commercially important plants. There are few scientific fields in which advances in understanding fundamental subjects, in this case understanding phylogenetic relationships, have had such unlucky shortcomings and real economic costs (Valleau, 2004; Mabberley, 2017).

Achieving consistency in the application of names requires that APG family and order names should not be altered unless (i) new results demonstrate non-monophyly or (ii) small families (often monogeneric/monotypic) are better treated as members of their sister families (e.g., expansion of Orobanchaceae to include several recently proposed small families with only brief histories of recognition/use). However, any such changes are likely to result in altered distributions of apomorphies for families/orders, and therefore, it is essential to balance these often-conflicting issues when contemplating changes.

The users of the APG classification are varied and cover numerous scientific fields beyond phylogenetics, systematics, and taxonomy, including agriculture, anatomy, biochemistry, conservation, developmental biology, ecology, floristics, horticulture, linguistics, paleobotany, pharmacognosy, physiology, and plant/animal/microbial interactions. Stability aids in literature searches by non-systematist users who are unfamiliar with the history of plant classification. Naturally, the APG system of classification is of great relevance to taxonomists and systematists, but changes in a classification affect numerous stakeholders across many other fields, for whom the fine details of phylogenetic relationships may be less important. It is nonetheless vital to carefully assess the broader impacts of any proposed changes (Mabberley, 2017). APG has a long history of using the advice of experts and results of phylogenetic studies produced by the phylogenetics community, particularly in the first two versions (1998, 2003), and this community-based content has been retained and predominates in the subsequent updates. However, it has not always been possible to reconcile some classifications proposed by experts with the principles that govern the APG classification. There are solutions to this dilemma (see below) that permit stability to be maintained while providing a more thorough and perhaps more synapomorphy-rich classification that many users will find valuable.

In the various iterations of APG, the number of families recognized has generally decreased over time as relationships have been clarified: 462 families in APG (1998) versus 416 families in APG IV (2016). APG V accepts 407 families. Reduction was especially obvious when the bracketed system in APG II (2003), which allowed users to choose between narrower or wider family limits, was abandoned in APG III (2009). This dual system was instituted so that experts could use narrower limits if they wished and still follow APG, while permitting other more general, non-systematist users to recognize more broadly defined families. The elimination of choice in APG III (2009) was due to many non-systematist users saying that they could not make these decisions based on their limited knowledge and wanting the APG compilers to abandon this option. In nearly all cases, the broader circumscriptions replaced the narrower ones, again with feedback from many users, and subsequently, the number of families recognized decreased: 457 families in APG II (2003) to 413 families in APG III (2009). Recognition of monofamilial orders was accepted since APG IV (2016), and these have been maintained in APG V. This adjustment in APG philosophy (since APG II, 2003) has been fundamental to maintaining stability of orders. In contrast to the number of families, the number of orders has increased, from 60 orders in APG II (2003) to 66 in APG V, largely due to an increase in those that are monofamilial.

With APG III (2009), stability became paramount, and APG IV (2016) and APG V (this paper) have involved only carefully assessed alterations of APG III, informed by the extensive nuclear and largely congruent plastid DNA evidence that provides the framework of APG V. In an assessment of the effects of APG on treatments of the flora of New Caledonia, Gotty & Pillon (2025) found that since 2012 (when the APG III, 2009, classification was implemented), family-level changes have been reduced to zero. This stability led these authors to conclude that "this classification can now be safely applied outside the world of scientific publication."

3 Operational Guidance and Clade-specific Issues

An alternative approach for specialists who believe that the broader family limits in APG are unsatisfactory is that taken for Asparagales, in which APG III used broad limits for Amaryllidaceae, Asparagaceae, and Xanthorrhoeaceae (now Asphodelaceae due to subsequent conservation; Klopper et al., 2018). For those interested in the finer details of relationships in these clades, Chase et al. (2009) published a subfamily classification of Amaryllidaceae, Asparagaceae, and Asphodelaceae (as Xanthorrhoeaceae). Chase et al. (2009) admitted that there were no morphological synapomorphies for some of the more broadly delimited families, but equally, there are no readily apparent non-DNA synapomorphies for many of the subfamilies of Chase et al. (2009). Thus, neither approach resulted in easily identifiable families/subfamilies. If family circumscriptions are going to be difficult to explain, then fewer, more broadly defined families may be easier for non-specialist users to remember and use than many, more difficult to explain families. Of course, many well-established families have had to be morphologically recircumscribed when DNA results showed them to include more morphologically divergent genera or genera formerly placed in other families (e.g., Acanthaceae, Gesneriaceae, Lamiaceae, and Scrophulariaceae). This resulted in families that are difficult to circumscribe morphologically in some cases, despite their

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monophyly, and the search for non-DNA synapomorphies remains a pressing need for many such clades in Asparagales, Lamiales, and other novel APG families without synapomorphies. It should also be noted that some non-DNA synapomorphies (such as phytochemistry or details of floral development, as opposed to general morphology) may be of little use in general family-level identification.

3.1 APG-based subfamily and suborder classifications

Classifications are conventions, and APG classifications are not exempt from this observation. APG was originally established to facilitate communication about phylogenetics of the angiosperms. Words and names are always subject to changes of meaning, but unnecessary modifications impede communication. An alternative to advocating the recognition of non-APG orders and families, which complements rather than conflicts with APG, would be to develop infra-ordinal and infra-familial classifications. These can provide specialists and interested general users with a framework reflecting ongoing discoveries of new clades requiring recognition, following the models of Colli-Silva et al. (2025) for Malvales (especially Malvaceae sensu APG), Larridon et al. (2021) for Cyperaceae, Buerki et al. (2021) for Sapindaceae, GPWG (2001, 2012, 2025) for Poaceae, LPWG (2017) and Koenen et al. (2020) for Fabaceae, and Pillon et al. (2024) for Oxalidales. In addition, monophyly of most subfamilies in several of the largest angiosperm families, including Asteraceae (Mandel et al., 2019; Zhang et al., 2021; Zhang et al., 2024), Orchidaceae (Zhang et al., 2023; Pérez-Escobar et al., 2024), Fabaceae (Koenen et al. 2020; Zhao et al., 2021), and Poaceae (Huang et al., 2022; Zhang et al. 2022; GPWG III, 2025), has been supported by large-scale nuclear gene analyses. Of course, recognition of distinct segregate families may draw the attention of more general users, but it often results in less attention for some groups because, for students and generalist users, the number of families becomes too large to grasp easily and causes confusion.

Smaller families are typically not covered in systematic botany and plant identification courses because the textbooks that cover these topics (e.g., Spichiger et al., 2004; Judd et al., 2015; Pandey & Kasana, 2021) cover roughly only a quarter of the APG families (i.e., never more than 100 families, mostly the larger ones). However, local flora courses or wild-flower guides often include some small families as a local component. As an example of the problem caused by splitting a family, in most courses on plant taxonomy, Boraginaceae s.l. (roughly 1800 species) are covered, but if split in up to 11 families (Luebert et al., 2016), not more than one of these families (eight with fewer than five genera) is likely to be included. Input from non-Boraginaceae specialists in an on-line survey (Christenhusz et al., 2015) revealed a preference for a single family for teaching purposes.

3.2 Condensation of small orders/families

A minor caveat to this principle of stability is when a monogeneric family or monofamilial order is found to be sister to another APG family or order and the two can be combined, especially when families share synapomorphies (e.g., combining Ixioliriaceae with Tecophilaeaceae and Frankeniaceae with Tamaricaceae; Backlund & Bremer, 1998). These actions are taken to increase the phylogenetic content of the system and reduce redundancy. This is not a new procedure, and it was also a tradition established in many past morphological classifications, for example, Cronquist (1981), to avoid assigning the rank of family to small groups if they can be accommodated in a related larger group. A higher taxon, for example, family or order, for a monospecific genus or monogeneric family was termed a "pseudon" by Corner (1981); it is thus a "pseudotaxon" or supererogatory taxon. In some cases, there is no choice in this matter (e.g., *Amborella* and *Ceratophyllum*), but these should be avoided. The shifting synapomorphies involved when combining orders are usually unclear at this point because many existing APG orders lack shared characteristics, but these are cases on which future studies could be focused.

3.3 Branching dates as guides for determining taxonomic ranks

The use of ages has been invoked by some authors as a criterion for determining an appropriate rank for clades and has been widely discussed in the taxonomic literature. The use of "time or temporal banding" to standardize ranks was supported in some cases by Vences et al. (2013) and for more general use by Avise & Mitchell (2007) and Kraichak et al. (2017). Angiosperm families have been demonstrated to have evolved at many different times (Wikström et al., 2001; Magallón et al. 2015, Zuntini et al., 2024), and any attempt to equalize the ages of families and orders recognized would result in massive, chaotic changes to the APG classification. Furthermore, as demonstrated in Magallón et al. (2015), different molecular dating methods have continued to provide divergent age estimates, and some models have led to widely differing age estimates for the same family. We consider attempts to equalize ages of families to be inappropriate (Lücking, 2019), and this procedure is not considered further here.

3.4 Conflicts in tree topologies

For purposes of classification, unambiguous, well-supported large-scale nuclear genomic trees are to be favored over plastid results if topologies conflict. As mentioned above, earlier versions of APG were primarily plastid-based by necessity, and general congruence among results using the three plant genomes has been noted and considered corroborative, albeit with some exceptions. Nuclear data sets analyzing general relationships across angiosperms such as Zeng et al. (2014) based on a relatively small number of taxa (61) and genes (59) first identified possible incongruence between nuclear and plastid topologies, as have other more focused studies, such as the analysis of mitochondrial genes for rosids by Sun et al. (2015) and all angiosperms in Lin et al. (2025). The much larger sampling of taxa and genes afforded by the 1KP project (410 nuclear orthologs; One Thousand Plant Transcriptomes Initiative, 2019), angiosperm-wide target sequence capture analyses of the Plant and Fungal Trees of Life (PAFTOL) project (Zuntini et al., 2024), and other studies of taxonomically less extensive datasets of nuclear genes (e.g., Stull et al., 2020; Guo et al., 2021; Alzate et al. 2024; Kates et al. 2024) have further reinforced the early indications of

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plastid versus nuclear incongruence at various levels across angiosperms. In addition to the discordance between organellar and nuclear trees, studies have also highlighted conflicting relationships identified with multiple nuclear genes (e.g., Escobar et al., 2011; Som, 2015; Sun et al., 2015).

Although noise, error, and paralogy probably contribute to some discordance between genes in the nuclear genomes, there has also been increasing support for biological causes of incongruence, including ILS, gene flow via hybridization/introgression, extensive rate heterogeneity, and possibly others. Genomic data have revealed multiple instances of deep incongruence in the angiosperms (Stull et al., 2023; Zuntini et al., 2024), although few of these phenomena affect the classification of families and orders, which are the focus of APG. The position of Ericales has been unresolved due to them likely being a product of ancient hybridization between ancestors of Cornales and the larger asterid clade (c. 120 Ma; Stull et al., 2020). Likewise, Guo et al. (2021) found evidence that ancient hybridization could account for the incongruent phylogenetic placement of Chloranthales and magnoliids in nuclear and plastid analyses (Table 1).

3.5 Plastid and nuclear trees

In APG V, we base alterations of APG IV (2016) on well-supported nuclear results, particularly if also supported by large analyses of plastome data (Li et al., 2021). The results on which circumscription of APG V orders and families are based have been generally consistent across phylogenetic methods and assumptions, and are parallel in coalescent, concatenated (combined), and most forms of weighted analyses. If extensive nuclear gene-tree incongruence has been detected, such that the position of a group is unresolved or only weakly supported, then we indicate uncertainty of result by considering them as "unplaced" to order or major clade (e.g., fabids, asterids). Likewise, if a set of relationships varies depending on the method used, we also indicate this as a polytomy for the groups concerned, unless noted in the section below on specific problematic parts of APG V (e.g., placement of Dasypogonaceae in Arecales, which has considerable gene-tree discordance at that node, that is, no portion of identical gene trees greater than 50% or, out of three possible quartet topologies, no more than two above one-third). Positions of some taxa in some plastid analyses should simply be discarded if they conflict with well-supported relationships from analyses of multiple nuclear genes. From an evolutionary standpoint, plastid trees are informative of, for example, the parentage of hybrids and phylogeographic population structure. However, if there is evidence of considerable nuclear gene-tree discordance (some of it perhaps agreeing with plastid results, but no congruent result in greater than 50% of nuclear gene trees) and/or if the source of discordance remains unclear, then, we propose to treat relationships as uncertain/preliminary, showing them as polytomies in the APG V summary tree (Fig. 1). We provide a summary of some examples (Table 1) and indicate the source of incongruence, if known. Many require further investigation before the source(s) of discordance can be clarified.

3.6 Reproducibility

Reproducibility in a classification is a desirable attribute (i.e., different authors reaching the same ideas of taxon delimitation based on the same data), but the idiosyncratic nature of plant classification makes this a difficult goal to achieve. The initial APG classification (1998) did not start with a clean slate, and all iterations of APG have been mixtures of historical concepts and new ideas about relationships. Many small family pairs, for example, Ancistrocladaceae–Dioncophyllaceae, Aextoxicaceae–Berberidopsidaceae, Curtisiaceae–Grubbiaceae, Asteropeiaceae–Physenaceae, and others, have been considered for merger in

Table 1 Examples of incongruence among trees for ancient clades

Citation	Clade(s)	Cause
Cai et al. (2021)	Malpighiales	ILS, hybridization
Sun et al. (2015)	Rosids	Hybridization, introgression
Stull et al. (2020) Larson et al. (2020)	Ericales	Hybridization, allopolyploidy, rapid radiation
Guo et al. (2021)	Magnoliids, Chloranthaceae, eudicots, monocots	Hybridization
Sun et al. (2024)	Malvaceae	Hybridization, allopolyploidy, ILS, homoploid introgression
Morales-Briones et al. (2021)	Amaranthaceae	ILS, rapid lineage diversification
Smith et al. (2015)	Caryophyllales	ILS, hybridization, polyploidy, gene duplication
Wang et al. (2021)	Caprifoliaceae	Hybridization
Joyce et al. (2023)	Sapindales	Polyploidy, hybridization
Feng et al. (2024)	Caryophyllaceae	Polyploidy
Wang et al. (2019)	Anacampserotaceae, Cactaceae, Didiereaceae, Montiaceae, Portulacaceae	Polyploidy
Walker et al. (2017)	Ancistrocladaceae, Droseraceae, Drosophyllaceae, Nepenthaceae	Polyploidy
Dong et al. (2022)	Oleaceae	Introgression, hybridization
Thomas et al. (2021)	Hydrostachyaceae, Cornales	Hybridization
Yu et al. (2025)	Sladeniaceae, Pentaphylacaceae	ILS

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the past and were judged not to share any characters, diagnosable or otherwise. Many of these small families have not been studied in sufficient detail beyond gross morphology to make such decisions, and until such time as they are more thoroughly investigated, their continued recognition in the APG classifications marks them as requiring more study. These four small-family pairs are not a complete list, and there are many more that could be considered. We have attempted to address these when we have had the opportunity, and undoubtedly, there are others that will be addressed in future iterations as their relationships become clearer and characters studied in greater detail (developmentally and genomically). Every such case of potential merger is unique and must be based on a combination of history, current knowledge of the genera/families in question, advice of specialists, and whether there are diagnostic characters or other, less obvious synapomorphies (e.g., phytochemistry, pollen morphology). However, given that this issue affects few families and is highly idiosyncratic (as discussed above), a general philosophy of when condensation should be invoked is impractical.

3.7 The parasite/holomycotroph problem

Parasites and holomycotrophs generally have been unsatisfactorily placed in the past based on organellar DNA analyses because the typical genes used are absent or highly modified in these groups. In the nuclear DNA sequencing results of Zuntini et al. (2024), these taxa were often frustratingly difficult to incorporate into matrices due to the low recovery of highly divergent genes. It can reasonably be expected that flowering plants with modified plastomes (i.e., parasites and holomycotrophs) will see reduced recovery of nuclear genes associated with organellar function, as is the case for many Angiosperms353 target genes (Johnson et al., 2019). The same applies to highly reduced aquatics, such as Ceratophyllaceae, Hydrostachyaceae, and Podostemaceae. Even with reasonable recovery, the highly divergent genes sequenced may possibly disrupt evidence of relationships with other families in these orders. We propose treating placements of several of these families as equivocal and position them only tentatively in orders, in some cases removing them from families in which they were formerly placed based solely on plastid and/or mitochondrial studies. In another case (Dioscoreales), we note the lack of resolution and support and propose to retain the APG IV classification, which includes families likely to prove non-monophyletic (see below).

4 Linear Classification and Poster

Orders in the summary tree (Fig. 1) and the list below are in a phylogenetic sequence if nuclear gene tree agreement and consistent results were obtained with different methods of analysis in the taxonomically well-sampled results of Zuntini et al. (2024). Listing of families in orders is in an alphabetical sequence in the Classification (Appendix A). In the Linear Classification (Appendix B), both orders and families in each order are in a phylogenetic sequence based on the results of Zuntini et al. (2024), with the convention of placing smaller families before the larger clade (or family) to which they are sister. A poster suitable for use in teaching is provided in the following link: https://www.tudelft.nl/en/hortus-botanicus/education/posters.

5 Discussion

Previous APG classifications, which were based almost solely on plastid DNA analyses, have been largely confirmed by recent studies with large-scale taxonomically informed sampling of hundreds of nuclear genes. As with other iterations of APG, we intend APG V to serve as a framework for communication not only among specialists but also more general users across diverse areas of biology in the broadest sense and other fields. Specialists may wish to propose alternative classifications for families and orders; we do not intend that APG should impede such proposals and discussions. Indeed, other classifications should be published and used as needed. However, we advocate that these should refer to the APG V classification so that interested general readers can follow the discussions and controversies. As stated above, the current composition of families and orders in APG V will remain unchanged unless an APG family or order is shown not to be monophyletic, with the caveat noted above for condensation of small families with their close relatives. However, we note that the relative arrangement of orders, especially in the core eudicots, is susceptible to change. Familial relationships in orders are also likely to change as new evidence emerges, and these are listed here (Appendix A) in an alphabetical sequence. In a few cases, we note below some remaining issues that may require modification of current circumscriptions when clarified.

6 Notes on Problematic Groups and Changes

1. Dioscoreaceae are a prime example of phylogenetic analysis problems caused by holomycotrophs, which mostly have reduced and altered plastid and mitochondrial genes/genomes (Shepeleva et al., 2020; Garrett et al., 2023). Gene recovery with target sequence capture has been lower than that for photosynthetic genera (Zuntini et al., 2024), most likely due to the divergent nature of these genes. Additionally, the nuclear genes that have been sequenced diverge considerably from those in the photosynthetic taxa (Soto Gomez et al., 2025). This introduces two problems: (i) uncertain placement and (ii) disruption of the relationships among the photosynthetic taxa because of putative long-branch attraction problems and uneven sampling of genes in the data sets. The species of *Afrothismia* Schltr. are especially problematic regarding (i) and (ii) above (Soto Gomez et al., 2025). Formerly treated as a member of Thismiaceae, *Afrothismia* was recently proposed as a family, Afrothismiaceae, by Cheek et al. (2024), who focused their work on the mitochondrial study of Lin et al. (2022). Based on the topology of that study, monogeneric Dioscoreaceae, Trichopodaceae, and Taccaceae were also recognized (*Stenomeris* Planch. was not included there). Cheek et al. (2024) listed the morphological differences between the dioscoreoid genera and argued that they are too different to be included in a single family. These views reflect an interpretation of what are "significant

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characters" and furthermore, the mitochondrial relationships shown by Lin et al. (2022), although important, are only part of the story. Support for interrelationships of these groups in Zuntini et al. (2024) and Soto Gomez et al. (2025) is poor, perhaps because of the divergent gene-sequence problems mentioned above. We propose to retain the classification of Dioscoreales from APG IV (2016), placing all thismioid genera in Dioscoreaceae s.l. until the relationships of these genera become better understood. We acknowledge that Burmanniaceae and Dioscoreaceae sensu APG IV (2016) appear non-monophyletic in some recent studies, depending on their circumscription. This type of approach was previously used for the families of Santalales in APG IV, which have at least been partially resolved at this point.

2. *Ixiolirion* Fisch. ex Herb. is supported as sister to Tecophilaeaceae in nuclear analyses, but its placement has been equivocal in some plastid analyses (e.g., Chase et al., 2006), and a sister relationship of these two in analyses of nuclear DNA is concordant with their similar morphology (leafy stems, bulbs, and flowers with mostly inferior ovaries). Tecophilaeaceae are therefore expanded here to include Ixioliriaceae. Morphologically, *Doryanthes* Correa (Doryanthaceae; two species) also fits well in Tecophilaeaceae, despite its large plant size, but without clear support from nuclear DNA analyses, we refrain from making this change.
3. In plastid trees (Barrett et al., 2016), Dasypogonaceae are a strongly supported sister of Arecaceae, but in the nuclear results (Zuntini et al., 2024), the position of the former is associated with gene-tree discordance. As in APG IV (2016), we include Dasypogonaceae in Arecales, but we acknowledge that this is a problematic placement that may yet need adjustment.
4. Piperales are another order in which holoparasites, that is, Hydnoraceae, have been problematic. Hydnoraceae have been shown in many analyses to be embedded in a broadly circumscribed Aristolochiaceae (reviewed in Naumann et al., 2013, and confirmed in Jost et al., 2021), when using genes from all three genomic compartments, but in Helmstetter et al. (2025), the favored topology for Hydnoraceae (coalescent analyses) was as a well-supported sister to the rest of Piperales. However, considerable gene-tree discordance was noted (35% of gene trees favored the embedded position in Aristolochiaceae), and some analyses (e.g., the concatenated matrix) supported the APG IV (2016) classification. The two genera of Hydnoraceae, *Hydnora* Thunb. and *Prosopanche* de Bary, share diagnostic characters with *Aristolochia* L. and *Asarum* L.: trimerous flowers with uniseriate perianth, inferior ovaries, and stamens fused with the style (gynostemium). We continue to treat Hydnoraceae here as members of Aristolochiaceae, but note that more research is required to clarify their relationships. *Lactoris* Phil. is embedded in Aristolochiaceae in the analyses of Zuntini et al. (2024) and Helmstetter et al. (2025), confirming their placement in APG IV (2016). If Lactoridaceae were to be recognized (as in Jost et al., 2021), then Asaraceae would also need to be recognized, breaking with a long tradition of their placement in Aristolochiaceae. We consider it preferable to keep Aristolochiaceae in their broad sense and use a subfamily classification to represent this diversity.
5. Relationships of Chloranthaceae have long been problematic, but they are well supported as sister to magnoliids in several recent nuclear DNA analyses (Guo et al., 2021; Zuntini et al., 2024), necessitating continued recognition of Chloranthales. A series of studies revealed incongruence between nuclear and plastid DNA topologies regarding the placement of magnoliids + Chloranthaceae relative to other members of the large mesangiosperm clade. Plastid trees place magnoliids + Chloranthaceae as sister to monocots plus eudicots, whereas nuclear genes usually recover monocots as sister to a large clade of magnoliids, Chloranthales, and eudicots (reviewed in Stull et al., 2023). Based on complete nuclear genome-sequence analyses, Guo et al. (2021) found evidence that ancient hybridization is the likely source of discordance over the position of Chloranthales + magnoliids relative to eudicots and monocots.
6. Although placement of Ceratophyllaceae was considered problematic in some early analyses with only a few genes or a few taxa, a series of both plastid and nuclear gene analyses has provided moderate to strong support for the family as sister to eudicots (e.g., Soltis et al., 2011; One Thousand Plant Transcriptomes Initiative, 2019; Yang et al., 2020; Guo et al., 2021), a placement that we follow here. However, Zuntini et al. (2024) recovered *Ceratophyllum* L. (but with little support) as sister to all angiosperms, except the ANA grade (Amborellales, Nymphaeales, and Austrobaileyales), a relationship also recovered in four nuclear gene analyses using 457 to 922 genes, but not in three others (Xiang et al., 2024). In analyses of nuclear data using model-based methods, Wang et al. (2025) placed *Ceratophyllum* as sister to the monocots, but changing the model in this study changed the placement, leading to little confidence in the monocots sister to *Ceratophyllum* result. The difficulty in placing *Ceratophyllum* may be due to early and rapid radiation of the mesangiosperms, as noted by Li et al. (2019). We include Ceratophyllaceae under the informal category of "ceratophyllids" in the classification (below) rather than including them in the eudicots and note that the position of *Ceratophyllum* requires further investigation before it can be classified confidently.
7. Apodanthaceae were firmly placed in Cucurbitales with plastid DNA (Li et al., 2021), as was the case in Filipowicz & Renner (2010) with mitochondrial (*matR*) and nuclear ribosomal DNA. In Zuntini et al. (2024), Apodanthaceae are positioned as sister to most if not all Malpighiales, but gene recovery was noted to be poor. However, in Alzate et al. (2024), who included more than 1000 nuclear loci, Apodanthaceae are well supported as sister to Rafflesiaceae, although with the sparse taxonomic sampling in this paper, it is unclear exactly where they go in Malpighiales. Historically, Apodanthaceae were considered members of Rafflesiaceae, with which they share many morphological characteristics, and we return them here to this position. It may be that lateral transfers of plastids and mitochondria from the host were responsible for

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their earlier placement in Cucurbitales (Filipowicz & Renner, 2010). Cai et al. (2025) used 2135 nuclear genes with several phylogenetic methods and found Rafflesiaceae (including *Apodanthes* Poit. and *Pilostyles* Guill.) to be variously placed with Euphorbiaceae, Peraceae, Putranjivaceae, and Pandaceae. Cai et al. (2025) suggested that parasitic taxa like Rafflesiaceae pose "a phylogenetic 'danger zone' where simulation suggests that current methods and genomic data will never provide a tidy phylogenetic resolution."

8. Peraceae (five genera, 106 species; POWO, 2025) were segregated from Euphorbiaceae in APG IV (2016) due to the interdigitation of Rafflesiaceae, but in Zuntini et al. (2024), Peraceae (represented by *Clutia* L. and *Chaetocarpus* Thwaites) are again sister to Euphorbiaceae, so we return the former to synonymy of the latter. As noted above, multiple nuclear loci place Rafflesiaceae (including Apodanthaceae) as sister to the rest of Malpighiales (Alzate et al., 2024).
9. Huaceae were formerly placed in Oxalidales (APG IV), but this position was not always well supported in plastid-only trees. In Li et al. (2021; plastomes only), they were a member of the larger COM (Celastrales/Oxalidales/Malpighiales) clade (as in APG IV, 2016), but the interrelationships of these orders were unclear. In Zuntini et al. (2024), Huaceae were closely related to Celastrales and Malpighiales, but not Oxalidales (see point 18 below); however, their specific placement is not well supported. Until relationships are better resolved, Huaceae should be considered a family unplaced to order. Recognition of Huales Doweld in Li et al. (2021), which was followed by Zuntini et al. (2024), is not followed here, due to the uncertain placement of Huaceae.
10. Relationships in Celastrales were studied by Simmons et al. (2023) using a combined analysis of DNA sequences and morphology; they recommended recognition of two families, Parnassiaceae (*Parnassia* L. and *Lepuropetalon* Elliott) and Celastraceae, but this classification is not followed here because Celastraceae s.l. have been a consistent feature since APG III (2009).
11. Santalales have continued to be problematic. In Zuntini et al. (2024), taxon sampling was limited, and previous results with a small number of plastid genes and nuclear 18S rDNA were not reproduced; some of the narrowly circumscribed families (summarized in Nickrent et al., 2010; updated in Su et al., 2015; and Nickrent et al., 2019) were not monophyletic. Olacaceae and Santalaceae sensu APG IV were known to be polyphyletic, but APG IV refrained from accepting the numerous small families proposed by Nickrent at al. (2010) because these were proposed without knowing their inter-relationships, which, if known, might make other options with fewer families viable. The families of Nickrent et al. (2010) were stated by the authors to be largely without obvious synapomorphies, and in such cases, fewer poorly circumscribed families would be preferable. Cauz-Santos et al. (2026) realigned the Su et al. (2015) matrix and found good support (83% bootstrap) for an Olacaceae clade (including Aptandraceae, Coulaceae, Octoknemaceae, and Ximeniaceae; all sensu Nickrent et al., 2010). Anderson et al. (2025), using Hyb-Seq nuclear data, found results similar to those in Cauz-Santos et al. (2026), but slightly less resolved. We leave Balanophoraceae s.l. unplaced in Santalales, despite the maximum likelihood results showing them to be well supported as embedded in Santalaceae (98% bootstrap; Cauz-Santos et al., 2026) because Anderson et al. (2025) found monophyletic Balanophoraceae s.l. not embedded in Santalaceae, but we note that there are considerable problems with gene recovery reported in the nuclear gene analyses (Anderson et al., 2025). This topic will need further consideration when more taxon sampling and better nuclear gene recovery have been achieved.
12. In Caryophyllales, Corbichoniaceae Thulin were proposed as a new family (Thulin et al., 2016) because of the morphological distinctiveness of *Corbichonia* Scop. relative to *Lophiocarpus* (Kunth) Miq. (Lophiocarpaceae). *Lophiocarpus* is wind-pollinated, and *Corbichonia* is insect-pollinated, syndromes common in larger families such as Caryophyllaceae and Amaranthaceae, to name a few, so this seems likely to be the cause of their dissimilarity and not the basis for separation. We continue to treat *Corbichonia* in Lophiocarpaceae despite the lack of morphological synapomorphies with *Lophiocarpus*.
13. Frankeniaceae comprise just *Frankenia* L., which is distinguished from Tamaricaceae (four genera) by opposite versus alternate leaves, respectively. They share small leaves with salt-excreting glands, basally clawed petals with a scale-like appendage or ligule on the upper side of the claw, secondary chemistry, and pollen morphology. They have always been considered a closely related family pair (e.g., Cronquist, 1981) and to reduce redundancy, we combine them under Frankeniaceae, which has nomenclatural priority.
14. In earlier iterations of APG, we failed to note that Schäferhoff et al. (2009) showed with two plastid markers that *Agdestis* Moç. & Sessé ex DC. is sister to *Sarcobatus* Nees, apart from Phytolaccaceae. A recent study using whole plastomes of Caryophyllales (Yao et al., 2019) confirmed these relationships, and Zuntini et al. (2024) found a similar result with nuclear DNA data. We must therefore either place Sarcobataceae (including *Agdestis*) in Phytolaccaceae or accept Agdestidaceae as an additional family. Here, we include *Agdestis* and *Sarcobatus* in Phytolaccaceae, despite the lack of clear synapomorphies linking *Sarcobatus* to Phytolaccaceae, because *Agdestis* sits comfortably in Phytolaccaceae on morphological grounds.
15. If the placement and identification of *Stixis* Lour. in Zuntini et al. (2024) can be verified, then Stixaceae could be recognized as distinct from Resedaceae, but gene recovery for the *Stixis* sample was low. Therefore, we think that the detailed study of Zhao et al. (2015) based on five plastid markers plus the nuclear ribosomal ITS region is more reliable because it includes several samples of *Stixis* and *Borthwickia* W.W.Sm. Zhao et al. (2015) also included *Neothorelia* Gagnep., *Forchhammeria* Liebm., and *Tirania* Pierre, all of which are critical in delimitation of Resedaceae. These collectively (but not *Borthwickia*) were sister to Resedaceae. However, the position of *Forchhammeria* in Zhao et al. (2015) vis-à-vis Resedaceae and *Stixis*/*Neothorelia*/*Tirania* was poorly supported. Rather than recognizing Borthwickiaceae,

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Stixaceae, and perhaps a new family for *Forchhammeria*, an enlarged Resedaceae would be preferable. More nuclear gene sampling and morphological study of the component genera should be conducted before any changes are made.

16. As noted above, Swanepoel et al. (2020) showed that *Batis* P.Brown., the sole genus of Bataceae, was well supported as sister to *Azima* Lam. (Salvadoraceae) and this pair sister to *Salvadora*, on which basis, we enlarge Salvadoraceae to include *Batis*. These taxa also share many morphological (inflorescence) features, making this a more reasonable change. Sampling in Zuntini et al. (2024) was not adequate to assess these relationships, but the position of *Batis* in Salvadoraceae has been consistent in all other analyses.
17. The position of the newly described Tiganophytaceae (Swanepoel et al., 2020) has differed depending on the genome/loci sequenced. They are either sister to Salvadoraceae (including Bataceae; Swanepoel et al., 2020) or embedded with Bataceae in a paraphyletic Salvadoraceae (Zuntini et al., 2024). Not all genera of Salvadoraceae have been included; most notably, *Dobera* Raf. is missing, but unpublished analyses by the PAFTOL project at Royal Botanic Gardens, Kew, place *Dobera* as sister to *Salvadora* Garcin ex L. (O. Maurin, pers., comm.). Morphologically, the three families share many floral characteristics and could be united, but given what is known thus far, recognition of Tiganophytaceae is appropriate.
18. In rosids, plastid results (e.g., Li et al., 2021) recovered the malvids and fabids as sister clades, but mitochondrial and nuclear results demonstrated a more complicated evolutionary scenario. Sun et al. (2015) first provided evidence that deep reticulation may have occurred in the early divergence of rosid clades; members of the COM families as defined in the plastid analyses (Celastrales, Oxalidales, Malpighiales) do not form a clade in Zuntini et al. (2024), and all three fall in the malvids rather than the fabids (the plastid tree placement). The nuclear trees of Zeng et al. (2017) further called into question the circumscriptions of these two large clades. Other, more recent nuclear DNA trees (e.g., One Thousand Plant Transcriptomes Initiative, 2019; Yang et al., 2020; Xiang et al., 2024; Zuntini et al., 2024; Zhao et al., 2025) have indicated that fabids should be limited just to the nitrogen-fixing members, Cucurbitales, Rosales, Fagales, and Fabales. This result was also obtained in the large mitochondrial analysis of Lin et al. (2025). In the malvids, there remains a core, well-supported clade of Malvales, Brassicales, Sapindales, and Oxalidales, but nuclear and mitochondrial results also place with them Huerteales, Picramniales, Huaceae, Malpighiales, and Celastrales (Zuntini et al., 2024; Lin et al., 2025). These additional orders plus Huaceae could be included in an enlarged malvid clade, but placements of Crossosomatales, Geraniales, Myrtales, and Zygophyllales are less clear, appearing in some trees as sister to both the reconfigured fabids and expanded malvids. Apart from the placement of Huaceae, none of these altered placements affects family or ordinal circumscriptions of APG IV, and our summary tree (Fig. 1) reflects these altered relationships found in the nuclear-gene analyses.
19. A similar problem to point 18 above occurs with the positions of Caryophyllales and Santalales, which, in plastid results (Li et al., 2021), were included in the superasterid clade along with Berberidopsidales and the asterids, but in nuclear and mitochondrial results, it is unclear where Caryophyllales and Santalales should be placed. Berberidopsidales are still sister to the large asterid clade and thus can be included in the superasterid clade, but we consider Caryophyllales and Santalales unplaced in the larger informally named superrosids and superasterids. Again, none of this uncertainty in placements affects the familial or ordinal circumscriptions of APG IV.
20. In APG IV, Vitales and Saxifragales were included in the rosids and superrosids, respectively, reflecting their positions in the plastid results (Zhang et al., 2016). In the mitochondrial trees, they are sister orders placed outside both the rosids and asterids (Lin et al., 2025). However, in the nuclear trees (Xiang et al., 2024; Zuntini et al., 2024; Zhao et al., 2025), their positions are reversed relative to the plastid results, with Vitales and Saxifragales successively sister to the rosids. We therefore include both here in the superrosids, as close relatives of the rosids. As for the orders and families discussed in points 18 and 19, this does not affect circumscriptions in APG IV.
21. APG V continues to recognize Boraginaceae in the broadest sense, synonymous with the monofamilial order Boraginales (APG IV, 2016). The split of Boraginaceae into 11 or nine families was argued by Luebert et al. (2016) and Vasile et al. (2025), respectively, to be the most conservative and consistent (less disruptive) with past treatments of the group. In both Vasile et al. (2025) and Cohen et al. (2025), there were shown to be problems with both the families as circumscribed by Luebert et al. (2016) and placement of some relative to the others (e.g., the position of Namaceae is weakly supported). For most botanists and other users of APG, nearly all the segregate families of Luebert et al. (2016) are unfamiliar. For example, in the treatment of Boraginaceae for the flora of New Caledonia, Heine (1976) and Gotty & Pillon (2025) rejected segregate families, the former stating (translated from the French) that "The family is understood here in its traditional sense (A.P. De Candolle, Bentham & J.D. Hooker, Gürke & Engler); the dismemberments often proposed from R. Brown ex Dumortier (Cordiaceae, 1829) and by Lindley (Ehretiaceae, 1830) are not taken into consideration, given that they are of little theoretical interest and their acceptance would ruin the incontestable homogeneity of this group of gamopetalous phanerogams with very natural systematic and phylogenetic relationships that are easy to bring together in the original conception of the Boraginaceae family (Jussieu 1789)." We agree with these sentiments, despite their age, and hence continue to follow the traditional treatment of Boraginaceae in the broad sense outlined in APG IV. We acknowledge that a subfamilial classification for Boraginaceae would be useful to incorporate what is known about relationships of the proposed separate families.
22. Recent work continues to support placement of monogeneric Hydrostachyaceae in Cornales (Thomas

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et al. 2021), perhaps most closely related to Loasaceae (Xu et al., 2024) or sister to the rest of Cornales (Zuntini et al., 2024), but Thomas et al. (2021) identified the involvement of ancient reticulation or ILS as affecting the position of *Hydrostachys* Thouars.

23. The position of *Ficalhoa* Hiern with respect to *Sladenia* Kurz (Sladeniaceae) and Pentaphylacaceae was unresolved in Zuntini et al. (2024), but in Yu et al. (2025), the former is well supported as sister to Pentaphylacaceae in analyses of nuclear DNA data (but not in their plastid trees). A merger of these two families under the older name Pentaphylacaceae is thus appropriate, given the lack of clear morphological apomorphies separating *Sladenia* from *Ficalhoa* and other Pentaphylacaceae (provided in Yu et al., 2025). Previously, Gilg & Werdermann (1925) and Robson (1962) considered Sladenieae and Ficalhoeae as tribes of Pentaphylacaceae, and Weitzman et al. (2004) also included them in Pentaphylacaceae. These proposals are complemented by the nuclear genomic analyses.
24. The families related to Orobanchaceae have experienced an ever-shifting number and generic composition since this family was recircumscribed to include several former genera of traditionally recognized Scrophulariaceae. Several small new families forming a grade to crown Orobanchaceae have been proposed, including Mazaceae, Paulowniaceae, Phrymaceae, Rehmanniaceae, and Wightiaceae (Beardsley & Olmstead, 2002; Reveal, 2011, 2012; Liu et al., 2020; Xia et al., 2021). As data became available for additional genera (Xia et al., 2009), unexpected relationships have been discovered, and new families have been proposed. For example, *Wightia* Spreng. ex DC. had been included in Paulowniaceae on morphological grounds until it was sequenced and found not to be sister to *Paulownia* Siebold & Zucc., resulting in a new family, Wightiaceae, being described (Liu et al., 2020). Furthermore, Xia et al. (2019) proposed that *Wightia* may be of hybrid origin between Phrymaceae and Paulowniaceae, and should thus be considered *incertae sedis*. Another genus previously thought to be a member of Paulowniaceae, *Brandisia* Hook.f. & Thomson, has now been shown to be a member of Orobanchaceae (Xia et al., 2019). Material of *Shiuyinghua silvestrii* (Pamp. & Bonati) Paclt (described as a species of *Catalpa* Scop.) has not been available for sequencing, so it remains with *Paulownia*, where the specimen (in Florence, FI) fits morphologically (JW Byng, unpubl.). Given that morphology has failed to facilitate an accurate assessment of relationships among these genera, continuing instability of family circumscriptions is likely. Although the problems with placement and circumscription of Paulowniaceae present intriguing evolutionary problems, they represent threats to stability. Rehmanniaceae were already included in Orobanchaceae (APG IV, 2016; Xia et al., 2021), and we continue this trend and expand Orobanchaceae here to include Mazaceae, Paulowniaceae, Phrymaceae, and Wightiaceae, which will stabilize the circumscription of Orobanchaceae while the process of establishing generic phylogenetic relationships and erection of new subfamilies/tribes continues.
25. In APG IV (2016), *Peltanthera* Benth. was tentatively left unplaced in the clade with Gesneriaceae and Calceolariaceae. However, immediately after publication of APG IV, Molinari-Novoa (2016) published Peltantheraceae, but not based on any additional information or data. *Peltanthera* has usually been associated with Gesneriaceae (Oxelman et al., 1999; Clark et al., 2010), and Luna et al. (2019) found *Peltanthera* closer to Gesneriaceae than Calceolariaceae, but with poor support, unlike Refulio-Rodríguez & Olmstead (2014), who recovered *Peltanthera* as sister to Gesneriaceae + Calceolariaceae. The broader clade here is consistent in all recent phylogenetic studies, and therefore, the most stable solution is to unite all three into the single family Gesneriaceae, perhaps with a subfamily classification that includes Calceolarioideae Luerss. and Peltantheroideae Molinari in addition to Didymocarpoideae Arnott, Gesnerioideae Link, and Sanangoideae A.Weber, J.L.Clark & Mich.Möller (Weber et al., 2013). An enlarged Gesneriaceae are united by inflorescence morphology, the ultimate units of the inflorescence being pair-flowered cymes omnipresent in Calceolariaceae and almost all Gesneriaceae (Weber, 2013). However, these are not found in *Peltanthera*, which has axillary, many-flowered panicles, but not all Gesneriaceae have pair-flowered cymes, exceptions being *Sinningia* Nees (single-flowered) and *Loxonia* Jack and *Stauranthera* Benth. (single species of each having ordinary cymes;- Weber, 2013), so inclusion of *Peltanthera* and Calceolariaceae, the latter with pair-flowered cymes, would not add otherwise unrepresented morphological diversity to Gesneriaceae.
26. The positions of Escalloniaceae and Paracryphiaceae require more study. In Zuntini et al. (2024), Escalloniaceae formed a grade to Paracryphiaceae, but this was weakly supported, and gene-tree discordance exists at these nodes. We continue to recognize Paracryphiales and Escalloniales but note here that changes may be needed in the future.
27. Oncothecaceae are well supported as an isolated campanulid family (Stull et al., 2020; Zhang et al., 2020) excluded from the other orders, and thus, we recognize Oncothecales Doweld.
28. *Cassinopsis* Sond. has been transferred from Icacinaceae to Metteniusaceae, which illustrates plastid–nuclear discordance (Stull et al., 2020); nuclear analyses correlate well with micromorphological synapomorphies. *Cassinopsis* was morphologically out of place in Icacinaceae.
29. In the plastid results of Li et al. (2021) and others, Cardiopteridales were included in Aquifoliales, but here, they are accepted as a separate order because they are sister to the larger lamiid clade (Zhang et al., 2020; Zuntini et al., 2024). They include Stemonuraceae and Cardiopteridaceae.
30. Expanding monogeneric Helwingiaceae to include monogeneric Phyllonomaceae (both Aquifoliales in APG IV, 2016) would then have epiphyllous inflorescences as a potential synapomorphy. Merging this sister pair would decrease redundancy in the system and has been consistent with the plastid results of several studies (Savolainen et al., 2000b; Li et al. 2021) and the more recent nuclear results of Zuntini et al. (2024).

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31. Columelliaceae (including *Desfontainia* Ruiz & Pav.) are no longer part of Bruniales and are treated here as an unplaced family close to Dipsacales and Escalloniales (Zhang et al., 2020; Zuntini et al., 2024). If a position sister to both these orders can be established for Columelliaceae, then recognition of Columelliales Mart. would be logical. The name Columelliales has priority over Desfontainiales Takht., and although priority does not apply above the rank of family, we would prefer to use priority as the primary criterion in the case of multiple, rarely used ordinal names.
32. Bruniales (without Columelliaceae) are unresolved in the campanulids (Zuntini et al. (2024), along with Oncothecales and Metteniusales, and thus, they are no longer close to Dipsacales/Apiales, as in plastid analyses (Li et al., 2021).
33. In Dipsacales, Viburnaceae has been conserved over Adoxaceae of APG II, III, and IV (Turland et al., 2018). If *Adoxa* L. s.l. is included in this family, the name Viburnaceae is now to be used.

7 Taxa *incertae sedis*

APG IV (2016) listed seven taxa as unplaced (*incertae sedis*). These were:

1. *Atrichodendron tonkinense* Gagnep. was placed in Solanaceae based on a poorly preserved specimen. It is certainly not Solanaceae in morphological terms, but characters are unclear in the single known specimen, and it has never been sequenced. It therefore remains unplaced to family.
2. *Coptocheile macrorrhiza* Hoffmanns. is a synonym of *Sinningia bulbosa* (Ker.Gawl.) Wiehler (Gesneriaceae), a species from southeastern Brazil (POWO, 2025).
3. *Gumillea auriculata* Ruiz & Pav. might belong to Picramniaceae, as suggested by Shipunov et al. (2020) and Christenhusz et al. (2017). It was described and illustrated from Peru, but no specimens are known. It is generally similar in morphology to *Picramnia*, but the flowers are bisexual, and the illustration shows stipules, which resemble pseudostipules of some species of *Picramnia* and some Huerteales. However, the genus will remain unplaced to family because no specimens are known.
4. *Hirania rosea* Thulin was confirmed as Sapindaceae by Buerki et al. (2021).
5. *Keithia brasiliensis* Spreng. remains unplaced. Suitable material for DNA analysis remains to be found. It was described as having a single sepal, five petals, and eight stamens, which does not match characters needed for inclusion in Capparaceae as originally suggested.
6. *Poilanedora unijuga* Gagnep. is similar to *Keithia* (above), but there are known specimens. It has leaves with two pinnae and five petals, like some Zygophyllaceae. However, the herbarium specimens have not yet yielded usable DNA data (F. Forest, pers. comm.), so its position has yet to be confirmed.
7. *Rumphia amboinensis* L. has been shown most likely to be a synonym of *Canarium strictum* Roxb. (Burseraceae) by Mabberley (2016), who based this on the text and illustration of the original in Van Rheede's *Hortus malabaricus*.

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We thank the hundreds of people who commented on an earlier draft of the manuscript and classification via an on-line survey. We received 234 responses: 33% from Europe, 26% from Asia, 23% from South America, 13% from North America, 3% from Australasia/Oceania, and 3% from Africa. Respondents were associated with a university (62%), herbarium (32%), botanical garden (30%), or museum (8%) (note that these categories were not mutually exclusive). Overall, 62% of the respondents were more than 80% in agreement with the APG V classification, 14% agreed with 50%–80%, and only 7% agreed with less than 50%; 17% had no opinion. Most of the disagreements were focused on Boraginales and Gesneriaceae/Calceolariaceae/Peltantheraceae. Both issues are discussed at length above. We gratefully acknowledge a grant from the Calleva Foundation to the Royal Botanic Gardens, Kew, for supporting an in-person meeting in October 2022, at which the substance of this update was discussed.

Authors Contributions

Mark W. Chase, Pamela S. Soltis, and Douglas E. Soltis led the writing with major contributions from James W. Byng, Maarten J. M. Christenhusz, Michael F. Fay, Peter F. Stevens, David J. Mabberley, De-Zhu Li, and Hong Ma. Contributions on their areas of speciality and/or advance information on upcoming publications were provided by William J. Baker, Steven Dodsworth, Felix Forest, Olivier Maurin, Lisa Pokorny, Stephen A. Smith, and Alexandre. R. Zuntini. James W. Byng created Figure 1.

Conflicts of Interest

The authors declare no conflict of interest.

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Appendix A
APG V Classification

Classification of angiosperms (APG V)

Phylogenetically organized by orders arranged in a phylogenetic sequence; families alphabetically listed in orders. Families published since 2016 are included as bracketed synonyms if not accepted here.

*changed circumscription of a family or newly added families since APG IV (2016).

†newly added or recircumscribed orders since APG IV (2016).

‡new relationship of major clades (not affecting circumscription of orders).

Families with single quotation marks, for example, 'Dioscoreaceae,' are noted not to be monophyletic in the circumscription used here and await more definitive phylogenetic analyses before formal classification is settled.

Amborellales Melikyan et al.

Amborellaceae Pichon, *nom. cons.*

Nymphaeales Salisb. ex Bercht. & J.Presl

Cabombaceae Rich. ex A.Rich., *nom. cons.*
Hydatellaceae U.Hamann
Nymphaeaceae Salisb., *nom. cons.*

Austrobaileyales Takht. ex Reveal

Austrobaileyaceae Croizat, *nom. cons.*
Schisandraceae Blume, *nom. cons.*
Trimeniaceae Gibbs, *nom. cons.*

MAGNOLIIDS

‡Chloranthales Mart.

Chloranthaceae R.Br. ex Sims, *nom. cons.*

Canellales Cronquist

Canellaceae Mart., *nom. cons.*
Winteraceae R.Br. ex Lindl., *nom. cons.*

Piperales Bercht. & J.Presl

Aristolochiaceae Juss., *nom. cons*
Piperaceae Giseke, *nom. cons.*
Saururaceae Rich. ex T.Lestib., *nom. cons.*

Magnoliales Juss. ex Bercht. & J.Presl

Annonaceae Juss., *nom. cons.*
Degeneriaceae I.W.Bailey & A.C.Sm., *nom. cons.*
Eupomatiaceae Orb., *nom. cons.*
Himantandraceae Diels, *nom. cons.*
Magnoliaceae Juss., *nom. cons.*
Myristicaceae R.Br., *nom. cons.*

Laurales Juss. ex Bercht. & J.Presl

Atherospermataceae R.Br.
Calycanthaceae Lindl., *nom. cons.*
Gomortegaceae Reiche, *nom. cons.*
Hernandiaceae Blume, *nom. cons.*
Lauraceae Juss., *nom. cons.*
Monimiaceae Juss., *nom. cons.*
Siparunaceae Schodde

MONOCOTS

Acorales Mart.

Acoraceae Martinov

Alismatales R.Br. ex Bercht. & J.Presl

Alismataceae Vent., *nom. cons.*
Aponogetonaceae Planch., *nom. cons.*
Araceae Juss., *nom. cons.*
Butomaceae Mirb., *nom. cons.*
Cymodoceaceae Vines, *nom. cons.*
Hydrocharitaceae Juss., *nom. cons.*
Juncaginaceae Rich., *nom. cons.*
Maundiaceae Nakai
Posidoniaceae Vines, *nom. cons.*
Potamogetonaceae Bercht. & J.Presl, *nom. cons*
Ruppiaceae Horan., *nom. cons.*
Scheuchzeriaceae F.Rudolphi, *nom. cons.*
Tofieldiaceae Takht.
Zosteraceae Dumort., *nom. cons.*

Petrosaviales Takht.

Petrosaviaceae Hutch., *nom. cons.*

Dioscoreales Mart.

'Dioscoreaceae' R.Br., *nom. cons.* (including *pro tem.*

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Afrothismiaceae Cheek & Soto Gomez, Taccaceae Dumort. Thismiaceae J.G.Agardh.)
Burmanniaceae Blume, *nom. cons.*
Nartheciaceae Fr. ex Bjurzon

Pandanales R.Br. ex Bercht. & J.Presl

Cyclanthaceae Poit. ex A.Rich., *nom. cons.*
Pandanaceae R.Br., *nom. cons.*
Stemonaceae Caruel, *nom. cons.*
Triuridaceae Gardner, *nom. cons.*
Velloziaceae J.Agardh, *nom. cons.*

Liliales Perleb

Alstroemeriaceae Dumort., *nom. cons.*
Campynemataceae Dumort.
Colchicaceae DC., *nom. cons.*
Corsiaceae Becc., *nom. cons.*
Liliaceae Juss., *nom. cons.*
Melanthiaceae Batsch ex Borkh., *nom. cons.*
Petermanniaceae Hutch., *nom. cons.*
Philesiaceae Dumort., *nom. cons.*
Ripogonaceae Conran & Clifford
Smilacaceae Vent., *nom. cons.*

Asparagales Link

Amaryllidaceae J.St.-Hil., *nom. cons.*
Asparagaceae Juss., *nom. cons.*
Asphodelaceae Juss., *nom. cons.*
Asteliaceae Dumort.
Blandfordiaceae R.Dahlgren & Clifford
Boryaceae M.W.Chase et al.
Doryanthaceae R.Dahlgren & Clifford
Hypoxidaceae R.Br., *nom. cons.*
Iridaceae Juss., *nom. cons.*
Lanariaceae H.Huber ex R.Dahlgren
Orchidaceae Juss., *nom. cons.*
*Tecophilaeaceae Leyb., *nom. cons.* (including Ixioliriaceae Nakai)
Xeronemataceae M.W.Chase et al.

Arecales Bromhead

Arecaceae Bercht. & J.Presl, *nom. cons., nom. alt.* (= Palmae Juss., *nom. cons., nom. alt.*)
Dasypogonaceae Dumort.

Commelinales Mirb. ex Bercht. & J.Presl

Commelinaceae Mirb., *nom. cons.*
Haemodoraceae R.Br., *nom. cons.*
Hanguanaceae Airy Shaw
Philydraceae Link, *nom. cons.*
Pontederiaceae Kunth, *nom. cons.*

Zingiberales Griseb.

Cannaceae Juss., *nom. cons.*
Costaceae Nakai
Heliconiaceae Vines
Lowiaceae Ridl., *nom. cons.*
Marantaceae R.Br., *nom. cons.*
Musaceae Juss., *nom. cons.*
Strelitziaceae Hutch., *nom. cons.*
Zingiberaceae Martinov, *nom. cons.*

Poales Small

Bromeliaceae Juss., *nom. cons.*
Cyperaceae Juss., *nom. cons.*
Ecdeiocoleaceae D.W.Cutler & Airy Shaw
Eriocaulaceae Martinov, *nom. cons.*
Flagellariaceae Dumort., *nom. cons.*
Joinvilleaceae Toml. & A.C.Sm.
Juncaceae Juss., *nom. cons.*
Mayacaceae Kunth, *nom. cons.*
Poaceae Barnhart, *nom. cons., nom. alt.* (= Gramineae Juss., *nom. cons., nom. alt.*)
Rapateaceae Dumort., *nom. cons.*
Restionaceae R.Br., *nom. cons*
Thurniaceae Engl., *nom. cons.*
Typhaceae Juss., *nom. cons.*
Xyridaceae C.Agardh, *nom. cons.*

CERATOPHYLLIDS

†Ceratophyllales Link

Ceratophyllaceae Gray, *nom. cons.*

EUDICOTS

Ranunculales Juss. ex Bercht. & J.Presl

Berberidaceae Juss., *nom. cons.*
Circaeasteraceae Hutch., *nom. cons.*
Eupteleaceae K.Wilh., *nom. cons.*
Lardizabalaceae R.Br., *nom. cons.*
Menispermaceae Juss., *nom. cons.*
Papaveraceae Juss., *nom. cons.*
Ranunculaceae Juss., *nom. cons.*

Proteales Juss. ex Bercht. & J.Presl

Nelumbonaceae A.Rich., *nom. cons.*
Platanaceae T.Lestib., *nom. cons.*
Proteaceae Juss., *nom. cons.*
Sabiaceae Blume, *nom. cons.*

Trochodendrales Takht. ex Cronquist

Trochodendraceae Eichler, *nom. cons.*

Buxales Takht. ex Reveal

Buxaceae Dumort., *nom. cons.*

CORE EUDICOTS

Gunnerales Takht. ex Reveal

Gunneraceae Meisn., *nom. cons.*
Myrothamnaceae Nied., *nom. cons.*

Dilleniales DC. ex Bercht. & J.Presl

Dilleniaceae Salisb., *nom. cons.*

SUPERROSIDS

†Vitales Juss. ex Bercht. & J.Presl

Vitaceae Juss., *nom. cons.*

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†Saxifragales Bercht. & J.Presl

Altingiaceae Lindl., *nom. cons.*
Aphanopetalaceae Doweld
Cercidiphyllaceae Engl., *nom. cons.*
Crassulaceae J.St.-Hil., *nom. cons.*
Cynomoriaceae Endl. ex Lindl., *nom. cons.*
Daphniphyllaceae Müll.Arg., *nom. cons.*
Grossulariaceae DC., *nom. cons.*
Haloragaceae R.Br., *nom. cons.*
Hamamelidaceae R.Br., *nom. cons.*
Iteaceae J.Agardh, *nom. cons.*
Paeoniaceae Raf., *nom. cons.*
Penthoraceae Rydb. ex Britton, *nom. cons.*
Peridiscaceae Kuhlm., *nom. cons.*
Saxifragaceae Juss., *nom. cons.*
Tetracarpaeaceae Nakai

ROSIDS

[‡rosids not in fabid or malvid clades]

‡Crossosomatales Takht. ex Reveal

Aphloiaceae Takht.
Crossosomataceae Engl., *nom. cons.*
Geissolomataceae A.DC., *nom. cons.*
Guamatelaceae S.H.Oh & D.Potter
Stachyuraceae J.Agardh, *nom. cons.*
Staphyleaceae Martinov, *nom. cons.*
Strasburgeriaceae Tiegh., *nom. cons.*

‡Geraniales Juss. ex Bercht. & J.Presl

Francoaceae A.Juss., *nom. cons.*
Geraniaceae Juss., *nom. cons.*

‡Zygophyllales Link

Krameriaceae Dumort., *nom. cons.*
Zygophyllaceae R.Br., *nom. cons.*

‡Myrtales Juss. ex Bercht. & J.Presl

Alzateaceae S.A.Graham
Combretaceae R.Br., *nom. cons.*
Crypteroniaceae A.DC., *nom. cons.*
Lythraceae J.St.-Hil., *nom. cons.*
Melastomataceae Juss., *nom. cons.*
Myrtaceae Juss., *nom. cons.*
Onagraceae Juss., *nom. cons.*
Penaeaceae Sweet ex Guill., *nom. cons.*
Vochysiaceae A.St.-Hil., *nom. cons.*

‡FABIDS

Fagales Engl.

Betulaceae Gray, *nom. cons.*
Casuarinaceae R.Br., *nom. cons.*
Fagaceae Dumort., *nom. cons.*
Juglandaceae DC. ex Perleb, *nom. cons.*
Myricaceae Rich. ex Kunth, *nom. cons.*
Nothofagaceae Kuprian.
Ticodendraceae Gómez-Laur. & L.D.Gómez

Fabales Bromhead

Fabaceae Lindl., *nom. cons., nom. alt.* (= Leguminosae Juss., *nom. cons., nom. alt.*)
Polygalaceae Hoffmanns. & Link, *nom. cons.*
Quillajaceae D.Don
Surianaceae Arn., *nom. cons.*

†Cucurbitales Juss. ex Bercht. & J.Presl (excluding Apodanthaceae Tiegh. ex Takht.)

Anisophylleaceae Ridl.
Begoniaceae C.Agardh, *nom. cons.*
Coriariaceae DC., *nom. cons.*
Corynocarpaceae Engl., *nom. cons.*
Cucurbitaceae Juss., *nom. cons.*
Datiscaceae Dumort., *nom. cons.*
Tetramelaceae Airy Shaw

Rosales Bercht. & J.Presl

Barbeyaceae Rendle, *nom. cons.*
Cannabaceae Martinov, *nom. cons.*
Dirachmaceae Hutch.
Elaeagnaceae Juss., *nom. cons.*
Moraceae Gaudich., *nom. cons.*
Rhamnaceae Juss., *nom. cons.*
Rosaceae Juss., *nom. cons.*
Ulmaceae Mirb., *nom. cons.*
Urticaceae Juss., *nom. cons.*

‡MALVIDS

Picramniales Doweld

Picramniaceae Fernando & Quinn

†Celastrales Link

Celastraceae R.Br., *nom. cons.*
Lepidobotryaceae J.Léonard, *nom. cons.*

[unplaced to order]

Huaceae A.Chev.

‡Malpighiales Juss. ex Bercht. & J.Presl

Achariaceae Harms, *nom. cons.*
Balanopaceae Benth. & Hook.f., *nom. cons.*
Bonnetiaceae L.Beauvis. ex Nakai
Calophyllaceae J.Agardh
Caryocaraceae Voigt
Centroplacaceae Doweld & Reveal
Chrysobalanaceae R.Br., *nom. cons.*
Clusiaceae Lindl., *nom. cons., nom. alt.* (= Guttiferae Juss., *nom. cons., nom. alt.*)
Ctenolophonaceae Exell & Mendonça
Dichapetalaceae Baill., *nom. cons.*
Elatinaceae Dumort., *nom. cons.*
Erythroxylaceae Kunth, *nom. cons.*
*Euphorbiaceae Juss., *nom. cons.* (including Peraceae Klotzsch)
Euphroniaceae Marc.-Berti
Goupiaceae Miers

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Humiriaceae A.Juss., *nom. cons.*
Hypericaceae Juss., *nom. cons.*
Irvingiaceae Exell & Mendonça, *nom. cons.*
Ixonanthaceae Planch. ex Miq., *nom. cons.*
Lacistemataceae Mart., *nom. cons.*
Linaceae DC. ex Perleb, *nom. cons.*
Lophopyxidaceae H.Pfeiff.
Malpighiaceae Juss., *nom. cons.*
Ochnaceae DC., *nom. cons.*
Pandaceae Engl. & Gilg, *nom. cons.*
Passifloraceae Juss. ex Roussel, *nom. cons.*
Phyllanthaceae Martinov, *nom. cons.*
Picrodendraceae Small, *nom. cons.*
Podostemaceae Rich. ex Kunth, *nom. cons.*
Putranjivaceae Meisn.
*Rafflesiaceae Dumort., *nom. cons.* (including Apodanthaceae Tiegh. ex Takht.)
Rhizophoraceae Pers., *nom. cons.*
Salicaceae Mirb., *nom. cons.*
Trigoniaceae A.Juss., *nom. cons.*
Violaceae Batsch, *nom. cons.*

Huerteales Doweld

Dipentodontaceae Merr., *nom. cons.*
Gerrardinaceae M.H.Alford
Petenaeaceae Christenh. et al.
Tapisciaceae Takht.

‡Oxalidales Bercht. & J.Presl (excluding Huaceae A.Chev.)

Brunelliaceae Engl., *nom. cons.*
Cephalotaceae Dumort., *nom. cons.*
Connaraceae R.Br., *nom. cons.*
Cunoniaceae R.Br., *nom. cons.*
Elaeocarpaceae Juss., *nom. cons.*
Oxalidaceae R.Br., *nom. cons.*

Sapindales Juss. ex Bercht. & J. Presl

Anacardiaceae R.Br., *nom. cons.*
Biebersteiniaceae Schnizl.
Burseraceae Kunth, *nom. cons.*
Kirkiaceae Takht.
Meliaceae Juss., *nom. cons.*
Nitrariaceae Lindl.
Rutaceae Juss., *nom. cons.*
Sapindaceae Juss., *nom. cons.*
Simaroubaceae DC., *nom. cons.*

Brassicales Bromhead

Akaniaceae Stapf, *nom. cons.*
Brassicaceae Burnett, *nom. cons., nom. alt.* (= Cruciferae Juss., *nom. cons., nom. alt.*)
Capparaceae Juss., *nom. cons.*
Caricaceae Dumort., *nom. cons.*
Cleomaceae Bercht. & J.Presl
Emblingiaceae Airy Shaw
Gyrostemonaceae A.Juss., *nom. cons.*
Koeberliniaceae Engl., *nom. cons.*
Limnanthaceae R.Br., *nom. cons.*
Moringaceae Martinov, *nom. cons.*
Pentadiplandraceae Hutch. & Dalziel
Resedaceae Martinov, *nom. cons.* (including *Borthwickia* W.W.Sm., *Forchhammeria* Liebm., Stixaceae Doweld)
*Salvadoraceae Lindl., *nom. cons.* (including Bataceae Mart. ex Perleb, *nom. cons.*)
Setchellanthaceae Iltis
Tovariaceae Pax, *nom. cons.*
Tropaeolaceae Juss. ex DC., *nom. cons.*
*Tiganophytaceae Swanepoel et al.

Malvales Juss. ex Bercht. & J.Presl

Bixaceae Kunth, *nom. cons.*
Cistaceae Juss., *nom. cons.* (including *Pakaraimaea* Maguire & P.S.Ashton)
Cytinaceae A.Rich.
Dipterocarpaceae Blume, *nom. cons.*
Malvaceae Juss., *nom. cons.*
Muntingiaceae C.Bayer et al.
Neuradaceae Kostel., *nom. cons.*
Sarcolaenaceae Caruel, *nom. cons.*
Sphaerosepalaceae Tiegh. ex Bullock
Thymelaeaceae Juss., *nom. cons.*

[non-rosid, non-asterid orders]

‡Santalales R.Br. ex Bercht. & J.Presl

*Balanophoraceae Rich., *nom. cons.* (including Mystropetalaceae Hook.f.)
*Erythropalaceae Planch. ex Miq., *nom. cons.*
Loranthaceae Juss., *nom. cons.*
Misodendraceae J.Agardh, *nom. cons.*
*Olacaceae R.Br., *nom. cons.* (excluding Erythropalaceae Planch. ex Miq., Strombosiaceae Tiegh.; including Aptandraceae Miers, Coulaceae van Tieghem, Octoknemaceae Tiegh., *nom. cons.*, Ximeniaceae Horaninow).
Opiliaceae Valeton, *nom. cons.*
*Santalaceae R.Br., *nom. cons.* (including Amphorogynaceae Nickrent & Der, Cervantesiaceae Nickrent & Der, Comandraceae Nickrent & Der, Nanodeaceae Nickrent & Der, Thesiaceae Vest, Viscaceae Batsch).
Schoepfiaceae Blume
*Strombosiaceae Tiegh.

‡Caryophyllales Juss. ex Bercht. & J.Presl

Achatocarpaceae Heimerl, *nom. cons.*
Aizoaceae Martinov, *nom. cons.*
Amaranthaceae Juss., *nom. cons.*
Anacampserotaceae Eggli & Nyffeler
Ancistrocladaceae Planch. ex Walp., *nom. cons.*
Asteropeiaceae Takht. ex Reveal & Hoogland
Barbeuiaceae Nakai
Basellaceae Raf., *nom. cons.*
Cactaceae Juss., *nom. cons.*
Caryophyllaceae Juss., *nom. cons.*
Didiereaceae Radlk., *nom. cons.*
Dioncophyllaceae Airy Shaw, *nom. cons.*
Droseraceae Salisb., *nom. cons.*
Drosophyllaceae Chrtek et al.

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*Frankeniaceae Desv., *nom. cons.* (including Tamaricaceae Link, *nom. cons.*)
Gisekiaceae Nakai
Halophytaceae S.Soriano
Kewaceae Christenh.
Limeaceae Shipunov ex Reveal
Lophiocarpaceae Doweld & Reveal (including Corbichoniaceae Thulin)
Macarthuriaceae Christenh.
Microteaceae Schäferhoff & Borsch
Molluginaceae Bartl., *nom. cons.*
Montiaceae Raf.
Nepenthaceae Dumort, *nom. cons.*
Nyctaginaceae Juss., *nom. cons.*
Petiveriaceae C.Agardh
Physenaceae Takht.
*Phytolaccaceae R.Br., *nom. cons.* (including Agdestidaceae Nakai and Sarcobataceae Behnke)
Plumbaginaceae Juss., *nom. cons.*
Polygonaceae Juss., *nom. cons.*
Portulacaceae Juss., *nom. cons.*
Rhabdodendraceae Prance
Simmondsiaceae Tiegh.
Stegnospermataceae Nakai
Talinaceae Doweld

SUPERASTERIDS

Berberidopsidales Doweld

Aextoxicaceae Engl. & Gilg, *nom. cons.*
Berberidopsidaceae Takht.

ASTERIDS

Cornales Link

Cornaceae Bercht. & J.Presl, *nom. cons.*
Curtisiaceae Takht.
Grubbiaceae Endl. ex Meisn., *nom. cons.*
Hydrangeaceae Dumort., *nom. cons.*
Hydrostachyaceae Engl., *nom. cons.*
Loasaceae Juss., *nom. cons.*
Nyssaceae Juss. ex Dumort., *nom. cons.*

Ericales Bercht. & J.Presl

Actinidiaceae Gilg & Werderm., *nom. cons.*
Balsaminaceae A.Rich., *nom. cons.*
Clethraceae Klotzsch, *nom. cons.*
Cyrillaceae Lindl., *nom. cons.*
Diapensiaceae Lindl., *nom. cons.*
Ebenaceae Gürke, *nom. cons.*
Ericaceae Juss., *nom. cons.*
Fouquieriaceae DC., *nom. cons.*
Lecythidaceae A.Rich., *nom. cons.*
Marcgraviaceae Bercht. & J.Presl, *nom. cons.*
Mitrastemonaceae Makino, *nom. cons.*
*Pentaphylacaceae Engl., *nom. cons.* (including Sladeniaceae Airy Shaw)
Polemoniaceae Juss., *nom. cons.*
Primulaceae Batsch ex Borkh., *nom. cons.*
Roridulaceae Martinov, *nom. cons.*
Sapotaceae Juss., *nom. cons.*
Sarraceniaceae Dumort., *nom. cons.*
Styracaceae DC. & Spreng., *nom. cons.*
Symplocaceae Desf., *nom. cons.*
Tetrameristaceae Hutch.
Theaceae Mirb., *nom. cons.*

‡LAMIIDS

†Cardiopteridales Takht.

Cardiopteridaceae Blume, *nom. cons.*
Stemonuraceae Kårehed

Garryales Mart.

Eucommiaceae Engl., *nom. cons.*
Garryaceae Lindl., *nom. cons.*

†Aquifoliales Senft (excluding Cardiopteridaceae Blume, Stemonuraceae Kårehed)

Aquifoliaceae Bercht. & J.Presl, *nom. cons.*
*Helwingiaceae Decne. (including Phyllonomaceae Small)

†Icacinales Tiegh.

Icacinaceae Miers, *nom. cons.* (excluding *Cassinopsis* Sond., now in Metteniusaceae H.Karst. ex Schnizl.)

Vahliales Doweld

Vahliaceae Dandy

Solanales Juss. ex Bercht. & J.Presl

Convolvulaceae Juss., *nom. cons.*
Hydroleaceae R.Br.
Montiniaceae Nakai, *nom. cons.*
Solanaceae Juss., *nom. cons.*
Sphenocleaceae T.Baskerv., *nom. cons.*

Boraginales Juss. ex Bercht. & J.Presl

Boraginaceae Juss., *nom. cons.* (including Coldeniaceae J.S.Mill. & Gottschling, Namaceae Molinari)

Gentianales Juss. ex Bercht. & J.Presl

Apocynaceae Juss., *nom. cons.*
Gelsemiaceae L.Struwe & V.A.Albert
Gentianaceae Juss., *nom. cons.*
Loganiaceae R.Br. ex Mart., *nom. cons.*
Rubiaceae Juss., *nom. cons.*

Lamiales Bromhead

Acanthaceae Juss., *nom. cons.*
Bignoniaceae Juss., *nom. cons.*
Byblidaceae Domin, *nom. cons.*
Carlemanniaceae Airy Shaw
*Gesneriaceae Rich. & Juss., *nom. cons.* (including Calceolariaceae Olmstead, Peltantheraceae Molinari-Novoa)
Lamiaceae Martinov, *nom. cons., nom. alt.* (= Labiatae Juss., *nom. cons., nom. alt.*)
Lentibulariaceae Rich., *nom. cons.*
Linderniaceae Borsch et al.
Martyniaceae Horan., *nom. cons.*
Oleaceae Hoffmanns. & Link, *nom. cons.*
*Orobanchaceae Vent., *nom. cons.* (including Mazaceae

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Reveal, Paulowniaceae Nakai, Phrymaceae Schauer, *nom. cons.*, Rehmanniaceae G.Kunkel ex Reveal, Wightiaceae Bo Li et al.)
Pedaliaceae R.Br., *nom. cons.*
Plantaginaceae Juss., *nom. cons.*
Plocospermataceae Hutch.
Schlegeliaceae Reveal
Scrophulariaceae Juss., *nom. cons.*
Stilbaceae Kunth, *nom. cons.*
Tetrachondraceae Wettst.
Thomandersiaceae Sreem.
Verbenaceae J.St.Hil., *nom. cons.*

‡CAMPANULIDS

†Oncothecales Doweld

Oncothecaceae Kobuski ex Airy Shaw

†Metteniusales Takht.

Metteniusaceae H.Karst. ex Schnizl. (including *Cassinopsis* Sond.)

†Bruniales Dumort. (excluding Columelliaceae D.Don, *nom. cons*)

Bruniaceae R.Br. ex DC., *nom. cons.*

Asterales Lin

Alseuosmiaceae Airy Shaw
Argophyllaceae Takht.
Asteraceae Bercht. & J.Presl, *nom. cons., nom. alt.* (= Compositae Giseke, *nom. cons., nom. alt.*)
Campanulaceae Juss., *nom. cons.*
Calyceraceae R.Br. ex Rich., nom. cons.
Goodeniaceae R.Br., *nom. cons.*
Menyanthaceae Dumort., *nom. cons.*
Pentaphragmataceae J.Agardh, *nom. cons.*
Phellinaceae Takht.
Rousseaceae DC.
Stylidiaceae R.Br., *nom. cons.*

[asterid, but unplaced to order]

Columelliaceae D.Don, *nom. cons.*

Escalloniales Link

Escalloniaceae R.Br. ex Dumort., *nom. cons.*

Paracryphiales Takht. ex Reveal

Paracryphiaceae Airy Shaw

Dipsacales Juss. ex Bercht. & J.Presl

Caprifoliaceae Juss., *nom. cons.*
*Viburnaceae Raf., *nom. cons.* (Adoxaceae E.Mey, *nom. cons.*; *sensu* APG II, 2003)

Apiales Nakai

Apiaceae Lindl., *nom. cons., nom. alt.* (= Umbelliferae Juss., *nom. cons., nom. alt.*)
Araliaceae Juss., *nom. cons.*
Griseliniaceae Takht., *nom. cons.*
Myodocarpaceae Doweld
Pennantiaceae J.Agardh
Pittosporaceae R.Br., *nom. cons.*
Torricelliaceae Hu

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Appendix B

Linear classification of flowering plants (APG V)
Numbers in square brackets are those of APG IV (2016).

1. Amborellales Melikyan et al.

1. [1]. Amborellaceae Pichon, *nom. cons.*

2. Nymphaeales Salisb. ex Bercht. & J.Presl

2. [2]. Hydatellaceae U.Hamann
3. [3]. Cabombaceae Rich. ex A.Rich., *nom. cons.*
4. [4]. Nymphaeaceae Salisb., *nom. cons.*

3. Austrobaileyales Takht. ex Reveal

5. [5]. Austrobaileyaceae Croizat, *nom. cons.*
6. [6]. Trimeniaceae Gibbs, *nom. cons.*
7. [7]. Schisandraceae Blume, *nom. cons.*

MAGNOLIIDS

4. Chloranthales Mart.

8. [26]. Chloranthaceae R.Br. ex Sims, *nom. cons.*

5. Canellales Cronq.

9. [8]. Canellaceae Mart., *nom. cons.*
10. [9]. Winteraceae R.Br. ex Lindl., *nom. cons.*

6. Piperales Bercht. & J.Presl

11. [12]. Aristolochiaceae Juss., *nom. cons.*
12. [10]. Saururaceae Rich. ex T.Lestib., *nom. cons.*
13. [11]. Piperaceae Giseke, *nom. cons.*

7. Magnoliales Juss. ex Bercht. & J.Presl

14. [13]. Myristicaceae R.Br., *nom. cons.*
15. [15]. Degeneriaceae I.W.Bailey & A.C.Sm., *nom. cons.*
16. [16]. Himantandraceae Diels, *nom. cons.*
17. [14]. Magnoliaceae Juss., *nom. cons.*
18. [17]. Eupomatiaceae Orb., *nom. cons.*
19. [18]. Annonaceae Juss., *nom. cons.*

8. Laurales Juss. ex Bercht. & J.Presl

20. [19]. Calycanthaceae Lindl., *nom. cons.*
21. [20]. Siparunaceae Schodde
22. [21]. Gomortegaceae Reiche, *nom. cons.*
23. [22]. Atherospermataceae R.Br.
24. [23]. Hernandiaceae Blume, *nom. cons.*
25. [24]. Monimiaceae Juss., *nom. cons.*
26. [25]. Lauraceae Juss., *nom. cons.*

MONOCOTS

9. Acorales Mart.

27. [27]. Acoraceae Martinov

10. Alismatales R.Br. ex Bercht. & J.Presl

28. [29]. Tofieldiaceae Takht.
29. [28]. Araceae Juss., *nom. cons.*
30. [30]. Alismataceae Vent., *nom. cons.*
31. [31]. Butomaceae Mirb., *nom. cons.*
32. [32]. Hydrocharitaceae Juss., *nom. cons.*
33. [34]. Aponogetonaceae Planch., *nom. cons.*
34. [33]. Scheuchzeriaceae F.Rudolphi, *nom. cons.*
35. [35]. Juncaginaceae Rich., *nom. cons.*
36. [36]. Maundiaceae Nakai
37. [39]. Posidoniaceae Vines, *nom. cons.*
38. [40]. Ruppiaceae Horan., *nom. cons.*
39. [41]. Cymodoceaceae Vines, *nom. cons.*
40. [37]. Zosteraceae Dumort., *nom. cons.*
41. [38]. Potamogetonaceae Bercht. & J.Presl, *nom. cons.*

11. Petrosaviales Takht.

42. [42]. Petrosaviaceae Hutch., *nom. cons.*

12. Dioscoreales Mart.

43. [43]. Nartheciaceae Fr. ex Bjurzon
44. [44]. Burmanniaceae Blume, *nom. cons.*
45. [45]. 'Dioscoreaceae' R.Br., *nom. cons.* (including pro tem. Afrothismiaceae Cheek & Soto Gomez, Taccaceae Dumort., Thismiaceae J.G.Agardh)

13. Pandanales R.Br. ex Bercht. & J.Presl

46. [47]. Velloziaceae J.Agardh, *nom. cons.*
47. [46]. Triuridaceae Gardner, *nom. cons.*
48. [48]. Stemonaceae Caruel, *nom. cons.*
49. [49]. Cyclanthaceae Poit. ex A.Rich., *nom. cons.*
50. [50]. Pandanaceae R.Br., *nom. cons.*

14. Liliales Perleb

51. [51]. Campynemataceae Dumort.
52. [52]. Corsiaceae Becc., *nom. cons.*
53. [53]. Melanthiaceae Batsch ex Borkh., *nom. cons.*
54. [54]. Petermanniaceae Hutch, *nom. cons.*
55. [55]. Alstroemeriaceae Dumort., *nom. cons.*
56. [56]. Colchicaceae DC., *nom. cons.*
57. [57]. Philesiaceae Dumort., *nom. cons.*
58. [58]. Ripogonaceae Conran & Clifford
59. [59]. Smilacaceae Vent., *nom. cons.*
60. [60]. Liliaceae Juss., *nom. cons.*

15. Asparagales Link

61. [61]. Orchidaceae Juss., *nom. cons.*
62. [62]. Boryaceae M.W.Chase et al.
63. [63]. Blandfordiaceae R.Dahlgren & Clifford
64. [64]. Asteliaceae Dumort.
65. [65]. Lanariaceae H.Huber ex R.Dahlgren
66. [66]. Hypoxidaceae R.Br., *nom. cons.*
67. [67]. Doryanthaceae R.Dahlgren & Clifford
68. [69]. Tecophilaeaceae Leyb., *nom. cons.* (including Ixioliriaceae Nakai)
69. [70]. Iridaceae Juss., *nom. cons.*
70. [71]. Xeronemataceae M.W.Chase et al.

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71. [72]. Asphodelaceae Juss., *nom. cons. prop.*
72. [73]. Amaryllidaceae J.St.-Hil., *nom. cons.*
73. [74]. Asparagaceae Juss., *nom. cons.*

16. Arecales Bromhead

74. [75]. Dasypogonaceae Dumort.
75. [76]. Arecaceae Bercht. & J.Presl, *nom. cons., nom. alt.* (= Palmae Juss., *nom. cons., nom. alt.*)

17. Commelinales Mirb. ex Bercht. & J.Presl

76. [80]. Pontederiaceae Kunth, *nom. cons.*
77. [81]. Haemodoraceae R.Br., *nom. cons.*
78. [79]. Philydraceae Link, *nom. cons.*
79. [77]. Hanguanaceae Airy Shaw
80. [78]. Commelinaceae Mirb., *nom. cons.*

18. Zingiberales Griseb.

81. [85]. Musaceae Juss., *nom. cons.*
82. [83]. Lowiaceae Ridl., *nom. cons.*
83. [82]. Strelitziaceae Hutch., *nom. cons.*
84. [84]. Heliconiaceae Vines
85. [86]. Cannaceae Juss., *nom. cons.*
86. [87]. Marantaceae R.Br., *nom. cons.*
87. [88]. Costaceae Nakai
88. [89]. Zingiberaceae Martinov, *nom. cons.*

19. Poales Small

89. [90]. Typhaceae Juss., *nom. cons.*
90. [91]. Bromeliaceae Juss., *nom. cons.*
91. [92]. Rapateaceae Dumort., *nom. cons.*
92. [96]. Thurniaceae Engl., *nom. cons.*
93. [97]. Juncaceae Juss., *nom. cons.*
94. [98]. Cyperaceae Juss., *nom. cons.*
95. [95]. Mayacaceae Kunth, *nom. cons.*
96. [94]. Eriocaulaceae Martinov, *nom. cons.*
97. [93]. Xyridaceae C.Agardh, *nom. cons.*
98. [99]. Restionaceae R.Br., *nom. cons.*
99. [100]. Flagellariaceae Dumort., *nom. cons.*
100. [102]. Ecdeiocoleaceae D.W.Cutler & Airy Shaw
101. [101]. Joinvilleaceae Toml. & A.C.Sm.
102. [103]. Poaceae Barnhart, *nom. cons., nom. alt.* (= Gramineae Juss., *nom. cons., nom. alt.*)

CERATOPHYLLIDS

20. Ceratophyllales Link

103. [104]. Ceratophyllaceae Gray, *nom. cons.*

EUDICOTS

21. Ranunculales Juss. ex Bercht. & J.Presl

104. [105]. Eupteleaceae K.Wilh., *nom. cons.*
105. [106]. Papaveraceae Juss., *nom. cons.*
106. [107]. Circaeasteraceae Hutch., *nom. cons.*
107. [108]. Lardizabalaceae R.Br., *nom. cons.*
108. [109]. Menispermaceae Juss., *nom. cons.*
109. [110]. Berberidaceae Juss., *nom. cons.*
110. [111]. Ranunculaceae Juss., *nom. cons.*

22. Proteales Juss. ex Bercht. & J.Presl

111. [112]. Sabiaceae Blume, *nom. cons.*
112. [113]. Nelumbonaceae A.Rich., *nom. cons.*
113. [114]. Platanaceae T.Lestib., *nom. cons.*
114. [115]. Proteaceae Juss., *nom. cons.*

23. Trochodendrales Takht. ex Cronq.

115. [116]. Trochodendraceae Eichler, *nom. cons.*

24. Buxales Takht. ex Reveal

116. [117]. Buxaceae Dumort., *nom. cons.*

CORE EUDICOTS

25. Gunnerales Takht. ex Reveal

117. [118]. Myrothamnaceae Nied., *nom. cons.*
118. [119]. Gunneraceae Meisn., *nom. cons.*

26. Dilleniales DC. ex Bercht. & J.Presl

119. [120]. Dilleniaceae Salisb., *nom. cons.*

SUPERROSIDS

27. Vitales Juss. ex Bercht. & J.Presl

120. [136]. Vitaceae Juss., *nom. cons.*

28. Saxifragales Bercht. & J.Presl

121. [135]. Cynomoriaceae Endl. ex Lindl., *nom. cons.*
122. [121]. Peridiscaceae Kuhlm., *nom. cons.*
123. [122]. Paeoniaceae Raf., *nom. cons.*
124. [126]. Daphniphyllaceae Müll.Arg., *nom. cons.*
125. [125]. Cercidiphyllaceae Engl., *nom. cons.*
126. [123]. Altingiaceae Lindl., *nom. cons.*
127. [124]. Hamamelidaceae R.Br., *nom. cons.*
128. [127]. Iteaceae J.Agardh, *nom. cons.*
129. [128]. Grossulariaceae DC., *nom. cons.*
130. [129]. Saxifragaceae Juss., *nom. cons.*
131. [132]. Tetracarpaeaceae Nakai
132. [131]. Aphanopetalaceae Doweld
133. [133]. Penthoraceae Rydb. ex Britton, *nom. cons.*
134. [134]. Haloragaceae R.Br., *nom. cons.*
135. [130]. Crassulaceae J.St.-Hil., *nom. cons.*

ROSIDS

[rosids not in fabid or malvid clades]

29. Crossosomatales Takht. ex Reveal

136. [223]. Aphloiaceae Takht.
137. [224]. Geissolomataceae A.DC., *nom. cons.*
138. [225]. Strasburgeriaceae Tiegh., *nom. cons.*
139. [227]. Guamatelaceae S.H.Oh & D.Potter
140. [228]. Stachyuraceae J.Agardh, *nom. cons.*
141. [229]. Crossosomataceae Engl., *nom. cons.*
142. [226]. Staphyleaceae Martinov, *nom. cons.*

30. Geraniales Juss. ex Bercht. & J.Presl

143. [213]. Francoaceae A.Juss., *nom. cons.*
144. [212]. Geraniaceae Juss., *nom. cons.*

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31. Zygophyllales Link

145. [137]. Krameriaceae Dumort., *nom. cons.*
146. [138]. Zygophyllaceae R.Br., *nom. cons.*

32. Myrtales Juss. ex Bercht. & J.Presl

147. [214]. Combretaceae R.Br., *nom. cons.*
148. [215]. Lythraceae J.St.-Hil., *nom. cons.*
149. [216]. Onagraceae Juss., *nom. cons.*
150. [220]. Crypteroniaceae A.DC., *nom. cons.*
151. [221]. Alzateaceae S.A.Graham
152. [222]. Penaeaceae Sweet ex Guill., *nom. cons.*
153. [219]. Melastomataceae Juss., *nom. cons.*
154. [217]. Vochysiaceae A.St.-Hil., *nom. cons.*
155. [218]. Myrtaceae Juss., *nom. cons.*

FABIDS

33. Fagales Engl.

156. [152]. Nothofagaceae Kuprian.
157. [154]. Myricaceae Rich. ex Kunth, *nom. cons.*
158. [155]. Juglandaceae DC. ex Perleb, *nom. cons.*
159. [156]. Casuarinaceae R.Br., *nom. cons.*
160. [157]. Ticodendraceae Gómez-Laur. & L.D.Gómez
161. [158]. Betulaceae Gray, *nom. cons.*
162. [153]. Fagaceae Dumort., *nom. cons.*

34. Fabales Bromhead

163. [141]. Surianaceae Arn., *nom. cons.*
164. [142]. Polygalaceae Hoffmanns. & Link, *nom. cons.*
165. [139]. Quillajaceae D.Don
166. [140]. Fabaceae Lindl., *nom. cons., nom. alt.* (= Leguminosae Juss., *nom. cons., nom. alt.*)

35. Cucurbitales Juss. ex Bercht. & J.Presl (excluding Apodanthaceae Tiegh. ex Takht.)

167. [161]. Corynocarpaceae Engl., *nom. cons.*
168. [160]. Anisophylleaceae Ridl.
169. [162]. Coriariaceae DC., *nom. cons.*
170. [163]. Cucurbitaceae Juss., *nom. cons.*
171. [165]. Datiscaceae Dumort., *nom. cons.*
172. [164]. Tetramelaceae Airy Shaw
173. [166]. Begoniaceae C.Agardh, *nom. cons.*

36. Rosales Bercht. & J.Presl

174. [145]. Dirachmaceae Hutch.
175. [144]. Barbeyaceae Rendle, *nom. cons.*
176. [146]. Elaeagnaceae Juss., *nom. cons.*
177. [147]. Rhamnaceae Juss., *nom. cons.*
178. [148]. Ulmaceae Mirb., *nom. cons.*
179. [149]. Cannabaceae Martinov, *nom. cons.*
180. [150]. Moraceae Gaudich., *nom. cons.*
181. [151]. Urticaceae Juss., *nom. cons.*
182. [143]. Rosaceae Juss., *nom. cons.*

MALVIDS

37. Picramniales Doweld

183. [230]. Picramniaceae Fernando & Quinn

38. Celastrales Link

184. [167]. Lepidobotryaceae J.Léonard, *nom. cons.*
185. [168]. Celastraceae R.Br., *nom. cons.*

[unplaced to order]

186. [169]. Huaceae A.Chev.

40. Malpighiales Juss. ex Bercht. & J.Presl

187. [206]. Rafflesiaceae Dumort., *nom. cons.* (including Apodanthaceae Tiegh. ex Takht.)
188. [178]. Ctenolophonaceae Exell & Mendonça
189. [176]. Pandaceae Engl. & Gilg, *nom. cons.*
190. [198]. Humiriaceae A.Juss., *nom. cons.*
191. [177]. Irvingiaceae Exell & Mendonça, *nom. cons.*
192. [181]. Ochnaceae DC., *nom. cons.*
193. [182]. Bonnetiaceae L.Beauvis. ex Nakai
194. [183]. Clusiaceae Lindl., *nom. cons., nom. alt.* (= Guttiferae Juss., *nom. cons., nom. alt.*)
195. [184]. Calophyllaceae J.Agardh
196. [185]. Podostemaceae Rich. ex Kunth, *nom. cons.*
197. [186]. Hypericaceae Juss., *nom. cons.*
198. [187]. Caryocaraceae Voigt, *nom. cons.*
199. [188]. Lophopyxidaceae H.Pfeiff.
200. [189]. Putranjivaceae Meisn.
201. [190]. Centroplacaceae Doweld & Reveal
202. [191]. Elatinaceae Dumort., *nom. cons.*
203. [192]. Malpighiaceae Juss., *nom. cons.*
204. [208]. Linaceae DC. ex Perleb, *nom. cons.*
205. [209]. Ixonanthaceae Planch. ex Miq., *nom. cons.*
206. [179]. Rhizophoraceae Pers., *nom. cons.*
207. [180]. Erythroxylaceae Kunth, *nom. cons.*
208. [193]. Balanopaceae Benth. & Hook.f., *nom. cons.*
209. [194]. Trigoniaceae A.Juss., *nom. cons.*
210. [195]. Dichapetalaceae Baill., *nom. cons.*
211. [196]. Euphroniaceae Marc.-Berti
212. [197]. Chrysobalanaceae R.Br., *nom. cons.*
213. [201]. Goupiaceae Miers
214. [199]. Achariaceae Harms, *nom. cons.*
215. [203]. Lacistemataceae Mart., *nom. cons.*
216. [204]. Salicaceae Mirb., *nom. cons.*
217. [202]. Passifloraceae Juss. ex Roussel, *nom. cons.*
218. [200]. Violaceae Batsch, *nom. cons.*
219. [210]. Picrodendraceae Small, *nom. cons.*
220. [211]. Phyllanthaceae Martinov, *nom. cons.*
221. [207]. Euphorbiaceae Juss., *nom. cons.* (including Peraceae Klotzsch)

41. Huerteales Doweld

222. [231]. Gerrardinaceae M.H.Alford
223. [232]. Petenaeaceae Christenh. et al.
224. [233]. Tapisciaceae Takht.
225. [234]. Dipentodontaceae Merr., *nom. cons.*

39. Oxalidales Bercht. & J.Presl (excluding Huaceae A.Chev.)

226. [170]. Connaraceae R.Br., *nom. cons.*
227. [171]. Oxalidaceae R.Br., *nom. cons.*
228. [174]. Cephalotaceae Dumort., *nom. cons.*
229. [172]. Cunoniaceae R.Br., *nom. cons.*

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230. [175]. Brunelliaceae Engl., *nom. cons.*
231. [173]. Elaeocarpaceae Juss., *nom. cons.*

42. Sapindales Juss. ex Bercht. & J.Presl

232. [235]. Biebersteiniaceae Schnizl.
233. [236]. Nitrariaceae Lindl.
234. [240]. Sapindaceae Juss., *nom. cons.*
235. [237]. Kirkiaceae Takht.
236. [238]. Burseraceae Kunth, *nom. cons.*
237. [239]. Anacardiaceae R.Br., *nom. cons.*
238. [242]. Simaroubaceae DC., *nom. cons.*
239. [241]. Rutaceae Juss., *nom. cons.*
240. [243]. Meliaceae Juss., *nom. cons.*

43. Brassicales Bromhead

241. [256]. Moringaceae Martinov, *nom. cons.*
242. [257]. Caricaceae Dumort., *nom. cons.*
243. [254]. Akaniaceae Stapf, *nom. cons.*
244. [255]. Tropaeolaceae Juss. ex DC., *nom. cons.*
245. [259]. Setchellanthaceae Iltis
246. [258]. Limnanthaceae R.Br., *nom. cons.*
247. [—]. Tiganophytaceae Swanepoel et al.
248. [262]. Salvadoraceae Lindl., *nom. cons.* (including Bataceae Mart. ex Perleb, *nom. cons.*)
249. [260]. Koeberliniaceae Engl., *nom. cons.*
250. [263]. Emblingiaceae Airy Shaw
251. [265]. Pentadiplandraceae Hutch. & Dalziel
252. [264]. Tovariaceae Pax, *nom. cons.*
253. [266]. Gyrostemonaceae A.Juss., *nom. cons.*
254. [267]. Resedaceae Martinov, *nom. cons.* (including *Borthwickia* W.W.Sm., *Forchhammeria* Liebm., Stixaceae Doweld)
255. [268]. Capparaceae Juss., *nom. cons.*
256. [269]. Cleomaceae Bercht. & J.Presl
257. [270]. Brassicaceae Burnett, *nom. cons.* (= Cruciferae Juss., *nom. cons.*)

44. Malvales Juss. ex Bercht. & J.Presl

258. [246]. Neuradaceae Kostel., *nom. cons.*
259. [244]. Cytinaceae A.Rich.
260. [249]. Thymelaeaceae Juss., *nom. cons.*
261. [245]. Muntingiaceae C.Bayer et al.
262. [248]. Sphaerosepalaceae Bullock
263. [250]. Bixaceae Kunth, *nom. cons.*
264. [251]. Cistaceae Juss., *nom. cons.* (including *Pakaraimaea* Maguire & P.S.Ashton)
265. [252]. Sarcolaenaceae Caruel, *nom. cons.*
266. [253]. Dipterocarpaceae Blume, *nom. cons.*
267. [247]. Malvaceae Juss., *nom. cons.*

[non-rosid, non-asterid orders]

45. Santalales R.Br. ex Bercht. & J.Presl

268. [—]. Strombosiaceae Tiegh.
269. [—]. Erythropalaceae Planch. ex Miq., *nom. cons.*
270. [273]. Olacaceae R.Br., *nom. cons.* (excluding Erythropalaceae Planch. ex Miq., Strombosiaceae Tiegh., including Aptandraceae Miers, Coulaceae van Tieghem, Octoknemaceae Tiegh., *nom. cons.*, Ximeniaceae Horaninow)
271. [274]. Opiliaceae Valeton, *nom. cons.*
272. [275]. Balanophoraceae Rich., *nom. cons.* (including Mystropetalaceae Hook.f.)
273. [276]. Santalaceae R.Br., *nom. cons.* (including Amphorogynaceae Nickrent & Der, Cervantesiaceae Nickrent & Der, Comandraceae Nickrent & Der, Nanodeaceae Nickrent & Der, Thesiaceae Vest, Viscaceae Batsch)
274. [277]. Misodendraceae J.Agardh, *nom. cons.*
275. [278]. Schoepfiaceae Blume
276. [279]. Loranthaceae Juss., *nom. cons.*

46. Caryophyllales Juss. ex Bercht. & J.Presl

277. [284]. Droseraceae Salisb., *nom. cons.*
278. [285]. Nepenthaceae Dumort, *nom. cons.*
279. [286]. Drosophyllaceae Chrtek et al.
280. [287]. Dioncophyllaceae Airy Shaw, *nom. cons.*
281. [288]. Ancistrocladaceae Planch. ex Walp., *nom. cons.*
282. [280]. Frankeniaceae Desv., *nom. cons.* (including Tamaricaceae Link, *nom. cons.*)
283. [282]. Plumbaginaceae Juss., *nom. cons.*
284. [283]. Polygonaceae Juss., *nom. cons.*
285. [289]. Rhabdodendraceae Prance
286. [290]. Simmondsiaceae Tiegh.
287. [291]. Physenaceae Takht.
288. [292]. Asteropeiaceae Takht. ex Reveal & Hoogland
289. [294]. Microteaceae Schäferhoff & Borsch
290. [298]. Stegnospermataceae Nakai
291. [296]. Achatocarpaceae Heimerl, *nom. cons.*
292. [297]. Amaranthaceae Juss., *nom. cons.*
293. [295]. Caryophyllaceae Juss., *nom. cons.*
294. [299]. Limeaceae Shipunov ex Reveal
295. [293]. Macarthuriaceae Christenh.
296. [300]. Lophiocarpaceae Doweld & Reveal (including Corbichoniaceae Thulin)
297. [302]. Barbeuiaceae Nakai
298. [301]. Kewaceae Christenh.
299. [304]. Aizoaceae Martinov, *nom. cons.*
300. [303]. Gisekiaceae Nakai
301. [305]. Phytolaccaceae R.Br., *nom. cons.* (including Agdestidaceae Nakai and Sarcobataceae Behnke)
302. [306]. Petiveriaceae C.Agardh
303. [308]. Nyctaginaceae Juss., *nom. cons.*
304. [309]. Molluginaceae Bartl., *nom. cons.*
305. [313]. Halophytaceae S.Soriano
306. [310]. Montiaceae Raf.
307. [312]. Basellaceae Raf., *nom. cons.*
308. [311]. Didiereaceae Radlk., *nom. cons.*
309. [314]. Talinaceae Doweld
310. [316]. Anacampserotaceae Eggli & Nyffeler
311. [315]. Portulacaceae Juss., *nom. cons.*
312. [317]. Cactaceae Juss., *nom. cons.*

SUPERASTERIDS

47. Berberidopsidales Doweld

313. [271]. Aextoxicaceae Engl. & Gilg, *nom. cons.*
314. [272]. Berberidopsidaceae Takht.

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ASTERIDS

48. Cornales Link

315. [319]. Hydrostachyaceae Engl., *nom. cons.*
316. [322]. Curtisiaceae Takht.
317. [323]. Grubbiaceae Endl. ex Meisn., *nom. cons.*
318. [318]. Nyssaceae Juss. ex Dumort., *nom. cons.*
319. [324]. Cornaceae Bercht. & J.Presl, *nom. cons.*
320. [320]. Hydrangeaceae Dumort., *nom. cons.*
321. [321]. Loasaceae Juss., *nom. cons.*

49. Ericales Bercht. & J.Presl

322. [346]. Mitrastemonaceae Makino, *nom. cons.*
323. [327]. Tetrameristaceae Hutch.
324. [326]. Marcgraviaceae Bercht. & J.Presl, *nom. cons.*
325. [325]. Balsaminaceae A.Rich., *nom. cons.*
326. [330]. Lecythidaceae A.Rich., *nom. cons.*
327. [328]. Fouquieriaceae DC., *nom. cons.*
328. [329]. Polemoniaceae Juss., *nom. cons.*
329. [334]. Ebenaceae Gürke, *nom. cons.*
330. [335]. Primulaceae Batsch ex Borkh., *nom. cons.*
331. [332]. Pentaphylacaceae Engl., *nom. cons.*
332. [333]. Sapotaceae Juss., *nom. cons.*
333. [336]. Theaceae Mirb., *nom. cons.*
334. [337]. Symplocaceae Desf., *nom. cons.*
335. [338]. Diapensiaceae Lindl., *nom. cons.*
336. [339]. Styracaceae DC. & Spreng., *nom. cons.*
337. [340]. Sarraceniaceae Dumort., *nom. cons.*
338. [341]. Roridulaceae Martinov, *nom. cons.*
339. [342]. Actinidiaceae Gilg & Werderm., *nom. cons.*
340. [343]. Clethraceae Klotzsch, *nom. cons.*
341. [344]. Cyrillaceae Lindl., *nom. cons.*
342. [345]. Ericaceae Juss., *nom. cons.*

LAMIIDS

52. Cardiopteridales Takht.

343. [389]. Cardiopteridaceae Blume, *nom. cons.*
344. [388]. Stemonuraceae Kårehed

53. Garryales Mart.

345. [350]. Eucommiaceae Engl., *nom. cons.*
346. [351]. Garryaceae Lindl., *nom. cons.*

54. Aquifoliales Senft (excluding Cardiopteridaceae Blume, Stemonuraceae Kårehed)

347. [391]. Helwingiaceae Decne. (including Phyllonomaceae Small)
348. [392]. Aquifoliaceae Bercht. & J.Presl, *nom. cons.*

55. Icacinales Tiegh. (excluding *Cassinopsis* Sond., Oncothecaceae Kobuski ex Airy Shaw)

349. [348]. Icacinaceae Miers, *nom. cons.*

56. Vahliales Doweld

350. [358]. Vahliaceae Dandy

57. Solanales Juss. ex Bercht. & J.Presl

351. [361]. Montiniaceae Nakai, *nom. cons.*
352. [362]. Sphenocleaceae T.Baskerv., *nom. cons.*
353. [363]. Hydroleaceae R.Br.
354. [359]. Convolvulaceae Juss., *nom. cons.*
355. [360]. Solanaceae Juss., *nom. cons.*

58. Boraginales Juss. ex Bercht. & J.Presl

356. [357]. Boraginaceae Juss., *nom. cons.* (including Coldeniaceae J.S.Mill. & Gottschling, Namaceae Molinari)

59. Gentianales Juss. ex Bercht. & J.Presl

357. [355]. Gelsemiaceae L.Struwe & V.A.Albert
358. [353]. Gentianaceae Juss., *nom. cons.*
359. [354]. Loganiaceae R.Br. ex Mart., *nom. cons.*
360. [356]. Apocynaceae Juss., *nom. cons.*
361. [352]. Rubiaceae Juss., *nom. cons.*

60. Lamiales Bromhead

362. [364]. Plocospermataceae Hutch.
363. [365]. Carlemanniaceae Airy Shaw
364. [366]. Oleaceae Hoffmanns. & Link, *nom. cons.*
365. [367]. Tetrachondraceae Wettst.
366. [369]. Gesneriaceae Rich. & Juss., *nom. cons.* (including Calceolariaceae Olmstead, Peltantheraceae Molinari-Novoa)
367. [370]. Plantaginaceae Juss., *nom. cons.*
368. [374]. Byblidaceae Domin, *nom. cons.*
369. [373]. Linderniaceae Borsch et al.
370. [371]. Scrophulariaceae Juss., *nom. cons.*
371. [372]. Stilbaceae Kunth, *nom. cons.*
372. [379]. Lentibulariaceae Rich., *nom. cons.*
373. [381]. Thomandersiaceae Sreem.
374. [376]. Pedaliaceae R.Br., *nom. cons.*
375. [377]. Acanthaceae Juss., *nom. cons.*
376. [380]. Schlegeliaceae Reveal
377. [382]. Verbenaceae J.St.Hil., *nom. cons.*
378. [375]. Martyniaceae Horan., *nom. cons.*
379. [378]. Bignoniaceae Juss., *nom. cons.*
380. [387]. Orobanchaceae Vent., *nom. cons.* (including Mazaceae Reveal, Paulowniaceae Nakai, Phrymaceae Schauer, Rehmanniaceae G.Kunkel ex Reveal, Wightiaceae Bo Li et al.)
381. [383]. Lamiaceae Martinov, *nom. cons.* (= Labiatae Juss., *nom. cons.*)

CAMPANULIDS

50. Oncothecales Doweld

382. [347]. Oncothecaceae Kobuski ex Airy Shaw

51. Metteniusales Takht. (including *Cassinopsis* Sond.)

383. [349]. Metteniusaceae H.Karst. ex Schnizl

61. Bruniales Dumort. (excluding Columelliaceae D.Don)

384. [406]. Bruniaceae R.Br. ex DC., *nom. cons.*

62. Asterales Link

385. [393]. Rousseaceae DC.
386. [395]. Pentaphragmataceae J.Agardh, *nom. cons.*
387. [394]. Campanulaceae Juss., *nom. cons.*
388. [396]. Stylidiaceae R.Br., *nom. cons.*

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389. [398]. Phellinaceae Takht.
390. [397]. Alseuosmiaceae Airy Shaw
391. [399]. Argophyllaceae Takht.
392. [400]. Menyanthaceae Dumort., *nom. cons*.
393. [401]. Goodeniaceae R.Br., *nom. cons*.
394. [402]. Calyceraceae R.Br. ex Rich., *nom. cons*.
395. [403]. Asteraceae Bercht. & J.Presl, *nom. cons., nom. alt*. (= Compositae Giseke, *nom. cons., nom. alt*.)

[asterid, unplaced to order]

396. [405]. Columelliaceae D.Don, *nom. cons*.

63. Escalloniales Link

397. [404]. Escalloniaceae R.Br. ex Dumort., *nom. cons*.

64. Paracryphiales Takht. ex Reveal

398. [407]. Paracryphiaceae Airy Shaw

65. Dipsacales Juss. ex Bercht. & J.Presl

399. [408]. Viburnaceae Raf., *nom. cons* [Adoxaceae E.Mey, *nom. cons*.]
400. [409]. Caprifoliaceae Juss., *nom. cons*.

66. Apiales Nakai

401. [410]. Pennantiaceae J.Agardh
402. [411]. Torricelliaceae Hu
403. [412]. Griseliniaceae Takht., *nom. cons*.
404. [413]. Pittosporaceae R.Br., *nom. cons*.
405. [414]. Araliaceae Juss., *nom. cons*.
406. [415]. Myodocarpaceae Doweld
407. [416]. Apiaceae Lindl., *nom. cons., nom. alt*. (= Umbelliferae Juss., *nom. cons., nom. alt*.)

Incertae sedis

Atrichodendron Gagnep.
Gumillea Ruiz & Pav. (possibly Picramniales or Huerteales)
Keithia Spreng.
Poilanedora Gagnep. (possibly Zygophyllales)

Supplementary Material

The following supplementary material is available online for this article at http://onlinelibrary.wiley.com/doi/10.1111/jse.70096/suppinfo:
Data S1. APG V Survey.

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